MODERN BRICK CONSTRUCTION SYSTEMS

A Catalogue of Affordable Solutions Made in Rwanda

Construction costs starting from FRW 150.000 per m^2

New solutions for challenging slope conditions

Featuring 12-in-1 garden houses for medium-high density locations

Introducing the Swiss Cube Typology System

THIRD EDITION

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By Skat Consulting, Ltd.

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Confédération suisse
Confederazione Svizzera
Confederaziun svizra

PROECCO Promoting Off-Farm Employment through Climate Responsive Construction Material Production

skat Swiss Resource Centre and Consultancies for Development

FOREWORD

Ban Ki-moon, former Secretary General of the UN:

"Income generation is closely associated with housing; it includes payments to construction works and construction suppliers, as well as home-based activities, some of which are linked to the global chain of production."

"Housing makes a considerable contribution to the national economic development in a variety of ways, including increases in capital stock, fixed investment and savings. In addition, there are significant interactions with financial systems, through housing banks, mortgage schemes, interest rates and consumption of housing services."

Source: Foreword from Dr. Anna Kajumulo Tibaijuka's
Building Prosperity: Housing and Economic Development, 2009

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OVERVIEW

MODERN BRICK MULTIPLEXES CAN BE AFFORDABLE TO MEDIUM-INCOME EARNERS

LOCAL BUILDING MATERIALS CAN GENERATE VIABLE SOLUTIONS TO HOUSING SUPPLY CHALLENGES

Construction costs in Rwanda are higher than in most other countries in Africa. This is mainly due to its land-locked geographic position and the resulting high transportation costs of imported material, namely of steel and cement. Rwanda's abundant clay deposits are of excellent quality and the massive demand of the country's fast-growing cities are fertile grounds for the construction industry to produce and build with Modern Brick Technologies. For several years Rwandan SME's, with the support of the Swiss Agency for Development and Cooperation, have started to produce machine-made Modern Bricks that allow for the construction of smart and cost-effective buildings. These technologies have the potential to significantly reduce the cost of housing and construction and bring tens of thousands of jobs back to Rwanda that were lost to the foreign cement industry.

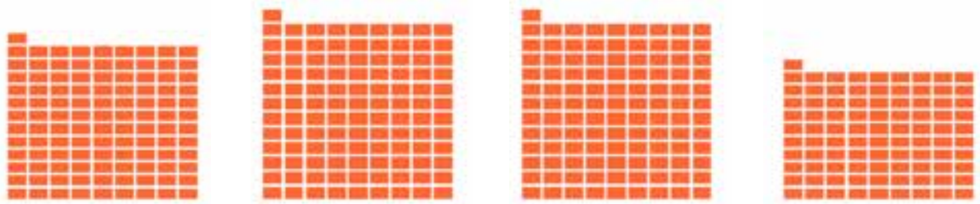
Stock of available Dwelling Units per Income Bracket in Kigali

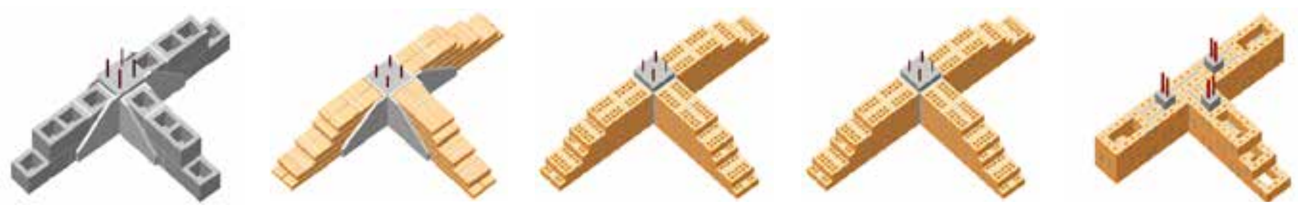
REASON Nº 1: MODERN BRICK WALLS ARE CHEAPER THAN TRADITIONAL BRICK OR CEMENT BLOCK WALLS

MODERN BRICKS ARE SEMI-INDUSTRIAL BRICKS PRODUCED BY RWANDAN 'SMEs'

BREAKDOWN OF MATERIAL COSTS
PER SQUARE METER OF WALL (INCLUDING TAXES)

	Hand-moulded Cement Block Wall	Traditional Brick Wall Outside facing English Bond	Modern Industrial Brick Wall Fully Facing English Bond	Modern Semi-Industrial Brick Wall Fully Facing English Bond	Modern Semi-Industrial Brick Wall Fully Facing Rowlock Bond
Cost	FRW 15.500	FRW 15.500	FRW 21.500	FRW 14.000	FRW 9.500
cement mortar	350 kg	206 kg	98 kg	98 kg	41 kg
bricks	0	121	121	121	101



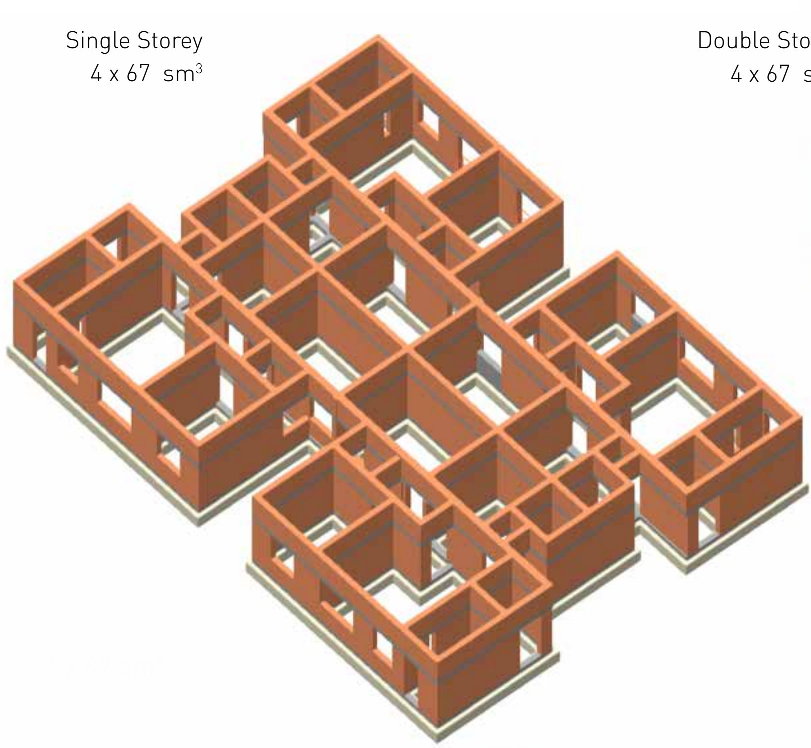


REASON Nº 2: MODERN BRICK MULTIPLEXES ARE CHEAPER THAN SINGLE-STORIED HOUSES

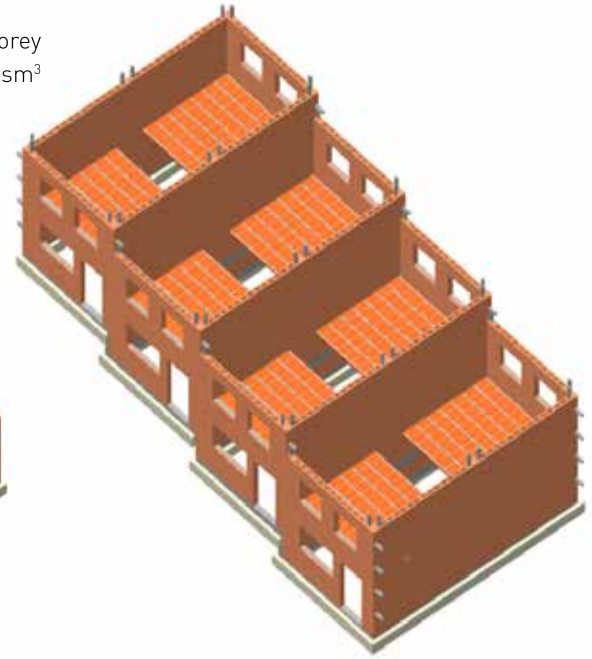
STRUCTURAL DESIGN DRIVES COST SAVINGS

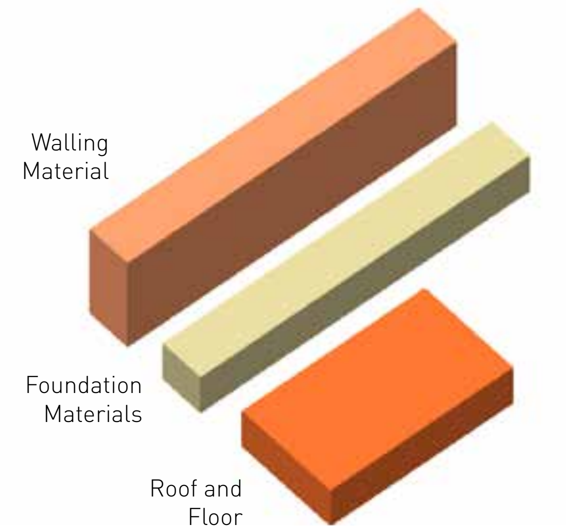
The Modern Brick Multiplex System is a standardised structural design for urban low-rise buildings, using RCC-reinforced Rowlock-Bond made of Modern Bricks. Its simple details are easy-to-apply and well-suited to medium-skilled masons who undergo a short training.

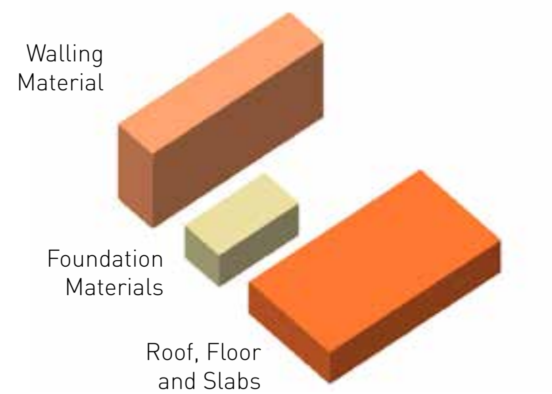
Single Storey
4 x 67 sm³



Double Storey
4 x 67 sm³







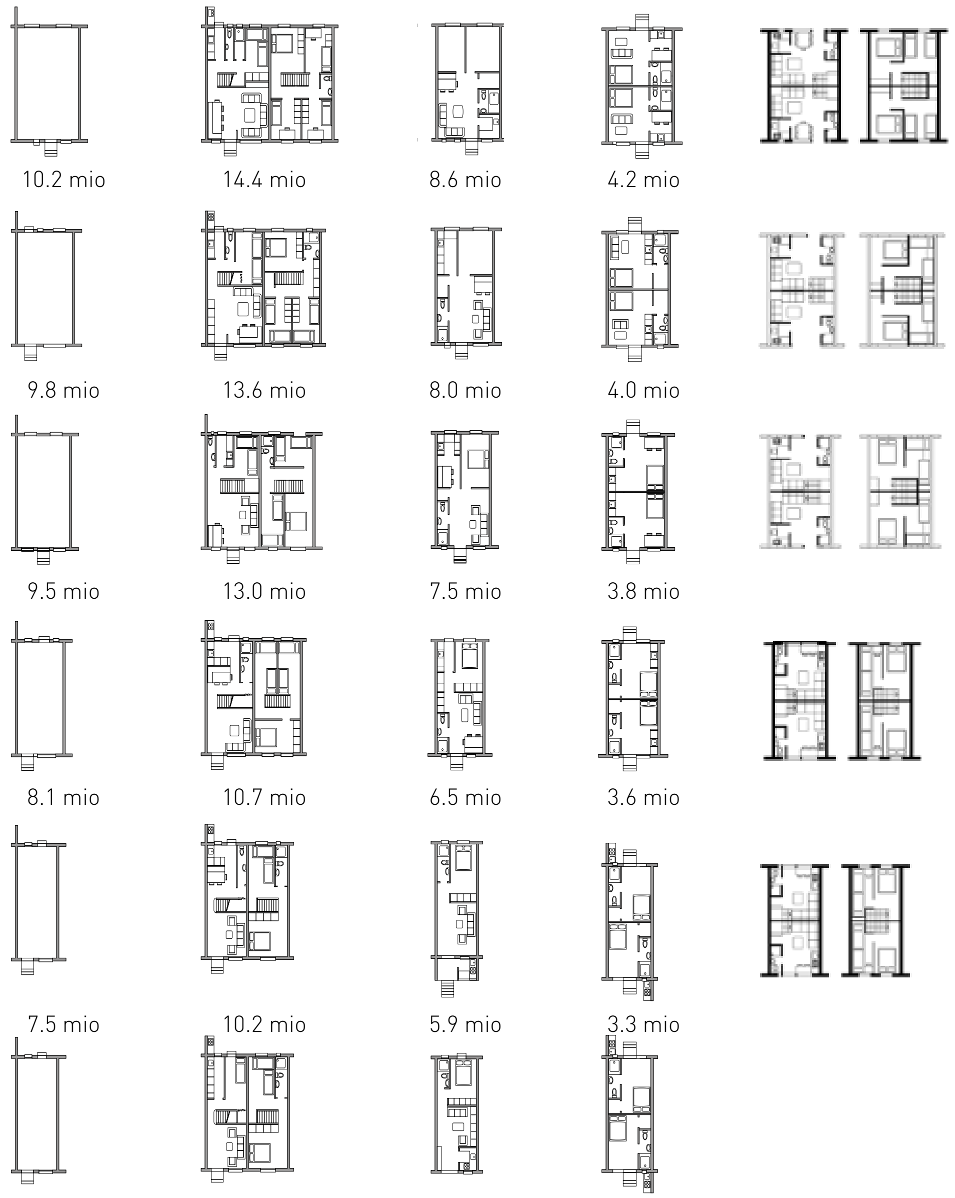
The Modern Brick Multiplex is particularly attractive for landlords who dream of a modern urban house and want to offer their middle-income tenants a modern and affordable house, apartment or studio.

- The S-size shell of a Modern Brick House Duplex (58m2), ready for interior works, costs FRW 6mio
- A Modern Brick Duplex with 3 bedrooms costs less than FRW 10 mio (5 bedrooms: FRW 15mio)
- Modern Brick Apartments can be built for less FRW 5mio, Studios for less than FRW 3mio
- A common plot can accommate up to 12 duplexes,14 simplexes or even 28 equipped studios!

REASON Nº 3: LANDOWNERS CAN COMPOSE THEIR BUILDING BY SELECTING SIZES/STANDARDS THEY CAN AFFORD

BUILDINGS ARE ASSEMBLED FROM A CATALOGUE OF SIMPLEX, DUPLEX AND TRIPLEX UNITS

Overview of modern brick duplex sizes and related costs* and options on how they can be used.

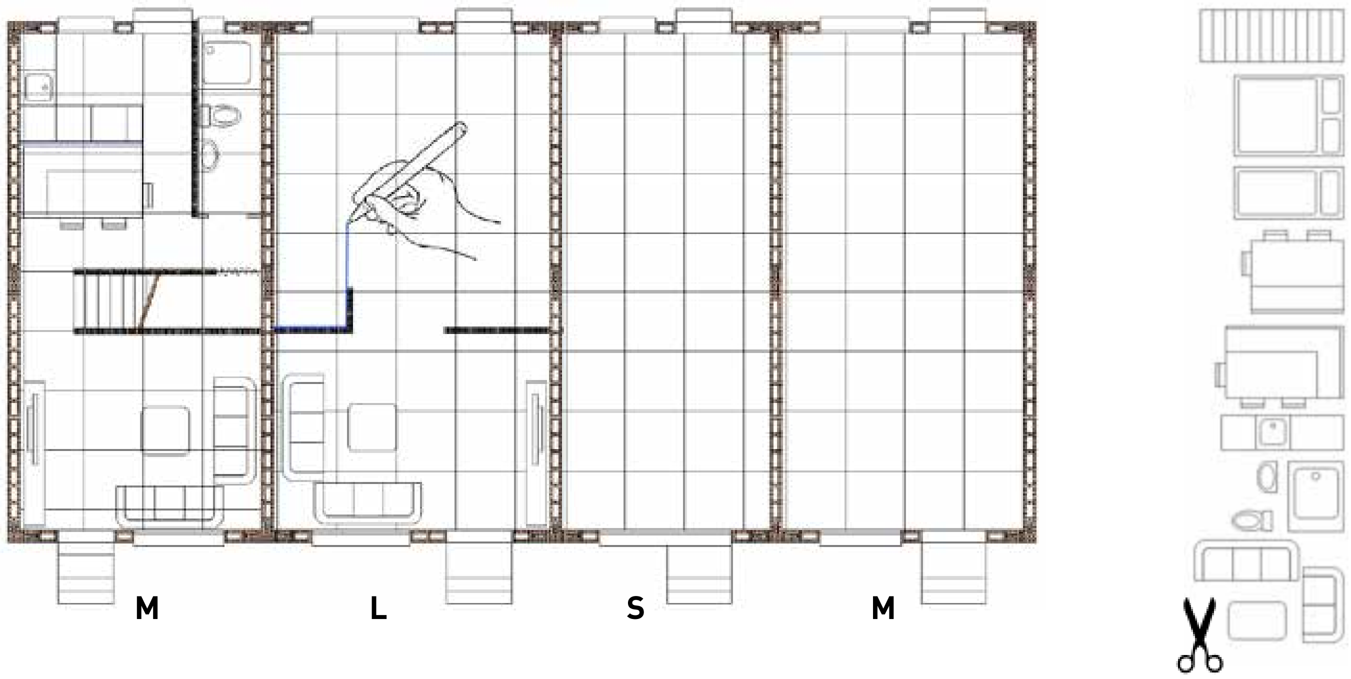


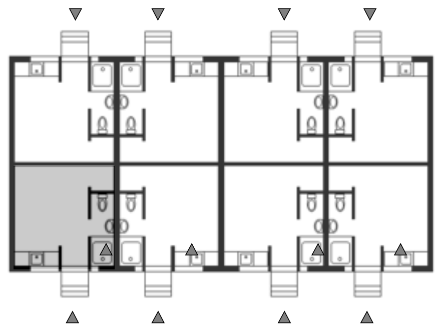
	SHELL	DUPLEX	APARTMENT	STUDIO	BACK-TO- BACK
XXL 5.25m x 8.3m 86m² or 2 x 43m²	10.2 mio	14.4 mio	8.6 mio	4.2 mio	
XL 4.9m x 8.3m 81m² or 2 x 41m²	9.8 mio	13.6 mio	8.0 mio	4.0 mio	
L 4.6m x 8.3m 77m² or 2 x 38m²	9.5 mio	13.0 mio	7.5 mio	3.8 mio	
M 4.1m x 8.3m 67m² or 2 x 33m²	8.1 mio	10.7 mio	6.5 mio	3.6 mio	
S 3.78m x 8.3m 63m² or 2 x 29m²	7.5 mio	10.2 mio	5.9 mio	3.3 mio	
XS 3.5m x 8.3m 58m² or 2 x 29m²	6.1 mio	9.2 mio	5.1 mio	2.9 mio	

*Calculated based on a simple plot in Kigali context that include all installations as well as tax and profit. Land excluded.

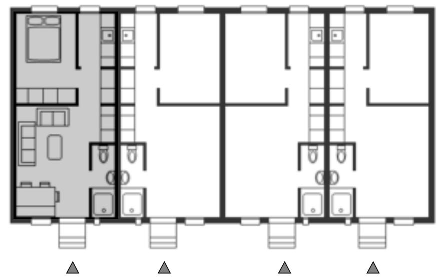
REASON Nº 4: THE INTERIOR OF A MODERN BRICK MULTIPLEX CAN BE CUSTOMISED AND CHANGED OVER TIME

ROOMS + WALLS CAN BE BUILT INCREMENTALLY IN RESPONSE TO EVOLVING NEEDS

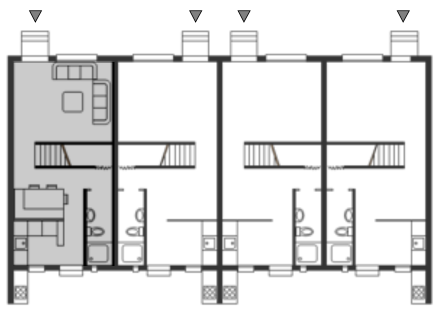




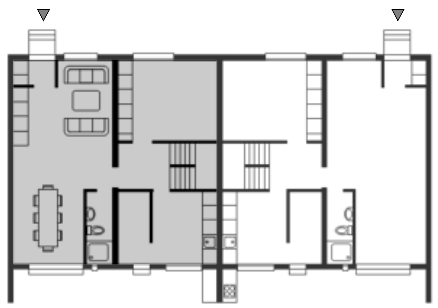
2016
Neighborhood Services: Water & Electricity only
Tenant Profile: Shop Employees, Taximoto Drivers
Tenant Income per HH: 100.000 FRW/month
16 Studios @ 35.000 FRW/month (8 per floor)



2022
Neighborhood Services: Pathways / drainage channels
Tenant Profile: Drivers, Artisans
Tenant income per HH: 200.000
8 Apartments @ 65.000/month (4 per floor)



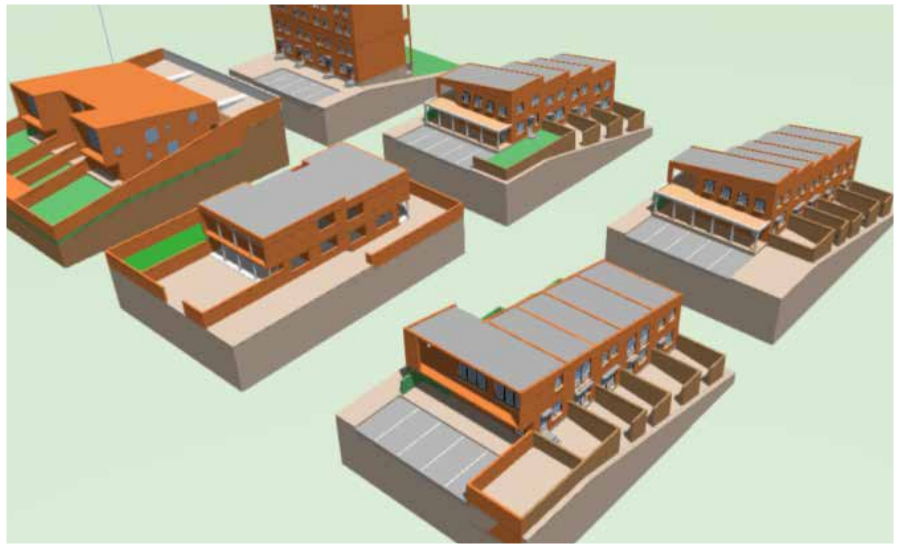
2028
Neighborhood Services: Asphalt Road
Tenant Profile: Clerks, Junior Engineers, Shopkeepers
Tenant income per HH: 400.000
4 duplexes @ 130.000/month



2036
Neighborhood Services: Public Transportation
Tenant Profile: Civil Servants, Entrepreneur
Tenant income per HH: 600.000
2 rowhouses @ 200.000/month

REASON Nº 5: MODERN URBAN BRICK MULTIPLEXES ARE SINGLE-PLOT SOLUTIONS FOR SMALL-SCALE LANDLORDS

ADAPTABLE TO ANY PLOT GEOMETRY, TOPOGRAPHY AND TENANT PROFILE



Adaptability

A standard plot size of 600 m² (20m x 30m) can accommodate a variety of duplex combinations (from rowhouses to apartment blocks), depending on the purchasing power of owners and tenants.

Optimisation

The flexible combination of units allows for optimized use of plot size, geometry and topography.

Sample 12-in-1 Building Configuration (left)

12-in-1 Stacked Duplex Solution
Estimated Construction Costs:
Starting from 96 million FRW

Sample 9-in-1 Building Configuration (below)

2 - Split-level back-to-back duplex
2 - Double Split-level duplex
1 - Split-level triplex (T*)
1 - Stacked split-level duplex
1 - Stacked Duplex
1 - Stacked Simplex
1 - Duplex
Estimated Construction Costs:
Starting from 92 million FRW

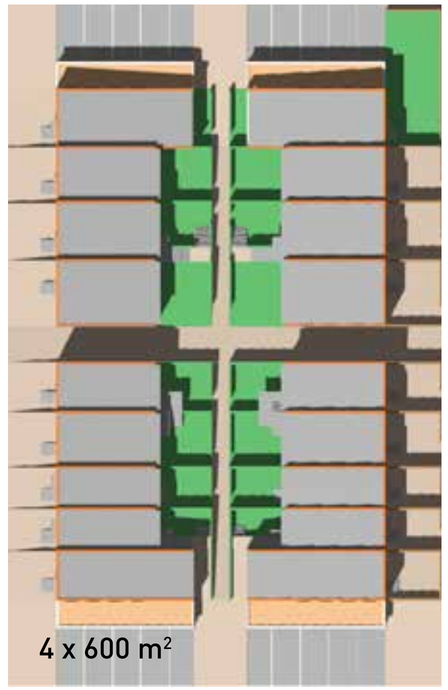
Axonometric of double split-level triplex duplex

REASON Nº 6: MODERN BRICK MULTIPLEXES ARE SUITABLE FOR SMALL NEIGHBORHOOD-LEVEL MICRO ESTATES

LANDOWNERS WHO JOIN HANDS TO UPGRADE THEIR ENVIRONS CAN FORMALISE THE BUILT ENVIRONMENT

78% of urban dwellings are in unplanned neighbourhoods and informally constructed due to the lack of affordable formal construction solutions. The formalisation of these neighborhoods becomes affordable through the Modern Brick duplex housing solution.

Upgrading begins at the plot level with individuals or small groups of landowners without disrupting existing tenancy patterns.
This method minimises the need for government intervention or expropriation.



Sample Micro-Sized Block Configuration (2 standard plots)

Two or three neighbours come together to create a small city block with the architectural and environmental qualities of a small estate with sufficient parking, walkways, private courtyards and a recognizable address.

Sample Micro-Sized Block Configuration (4 standard plots)

Neighbours organise themselves into cooperatives to create 20 Modern Brick Duplexes accommodating anywhere from 20 to 80 dwelling units, including shops, services and small private offices.

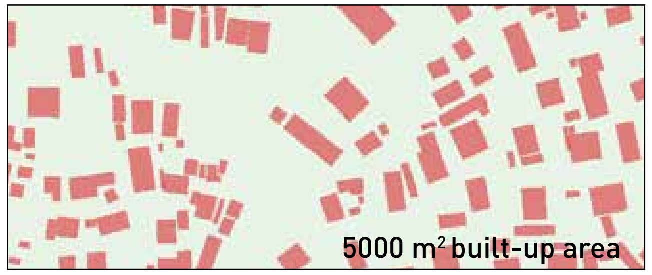
REASON Nº 7: LANDLORDS BUILDING MODERN URBAN BRICK MULTIPLEXES CAN HOUSE TWICE THE NUMBER OF TENANTS

MODERN BRICK MULTIPLEX HOUSES ARE SUITABLE FOR MANY DIFFERENT ZONING DESIGNATIONS

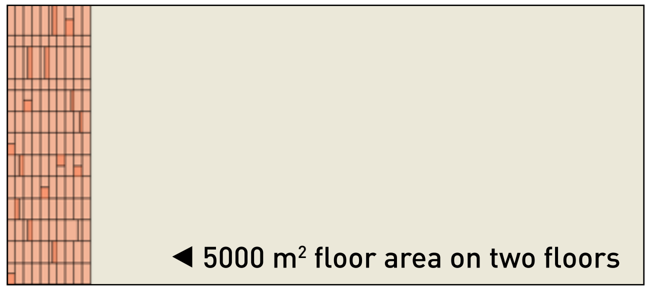
REASON Nº 8: SMART DENSIFICATION ALLOWS SMALL LANDOWNERS TO SELF-FINANCE NEIGHBORHOOD UPGRADING

LAND VALUE IN INFORMAL AREAS CAN BE UNLOCKED THROUGH COMPACT AFFORDABLE BUILDING SOLUTIONS

Irregular plot and building distribution
Typical Unplanned Settlement in Kigali
Built up area as shown totals 5000 m^2.



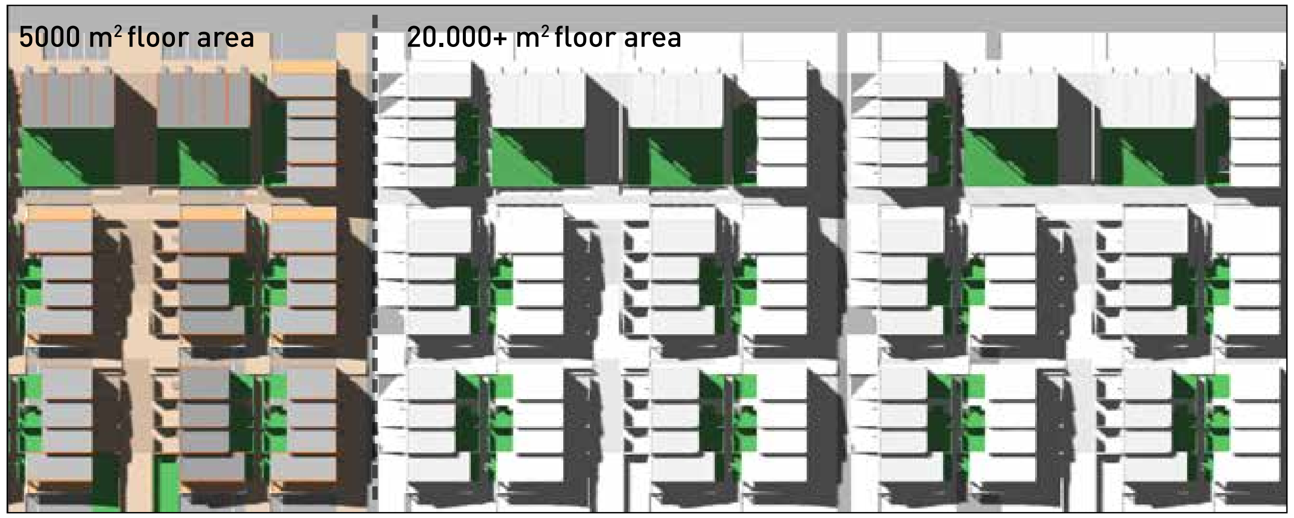
Existing building coverage is low
Study of building footprints reveals that less than 80% of land is occupied.



Storied houses can improve land use
Building an additional floor and compacting structures can free up land for development.

Compact development can unlock value of land
Available land can be sold to co-finance the in-situ resettlement of land owners and tenants into storied modern brick houses (left).

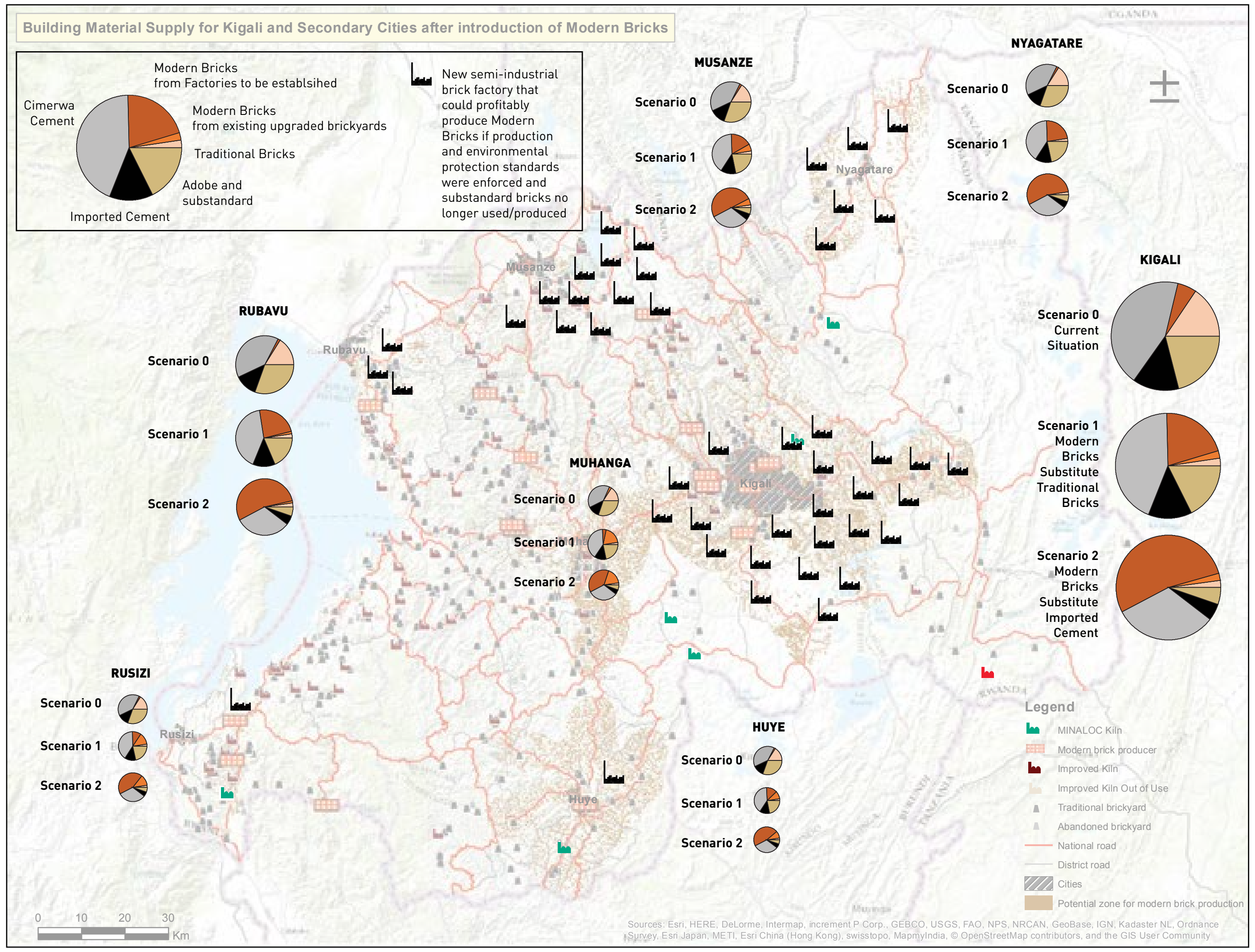
After self-financing, the floor area could be increased as per below.

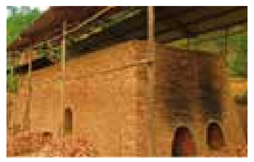


REASON Nº 9: MODERN BRICKS CAN BE PRODUCED BY LOCAL SMEs AND SUBSTITUTE IMPORTED CEMENT

PRODUCTION UNITS IN AND AROUND THE SECONDARY CITIES COULD RESPOND TO THE DEMAND FOR MATERIALS

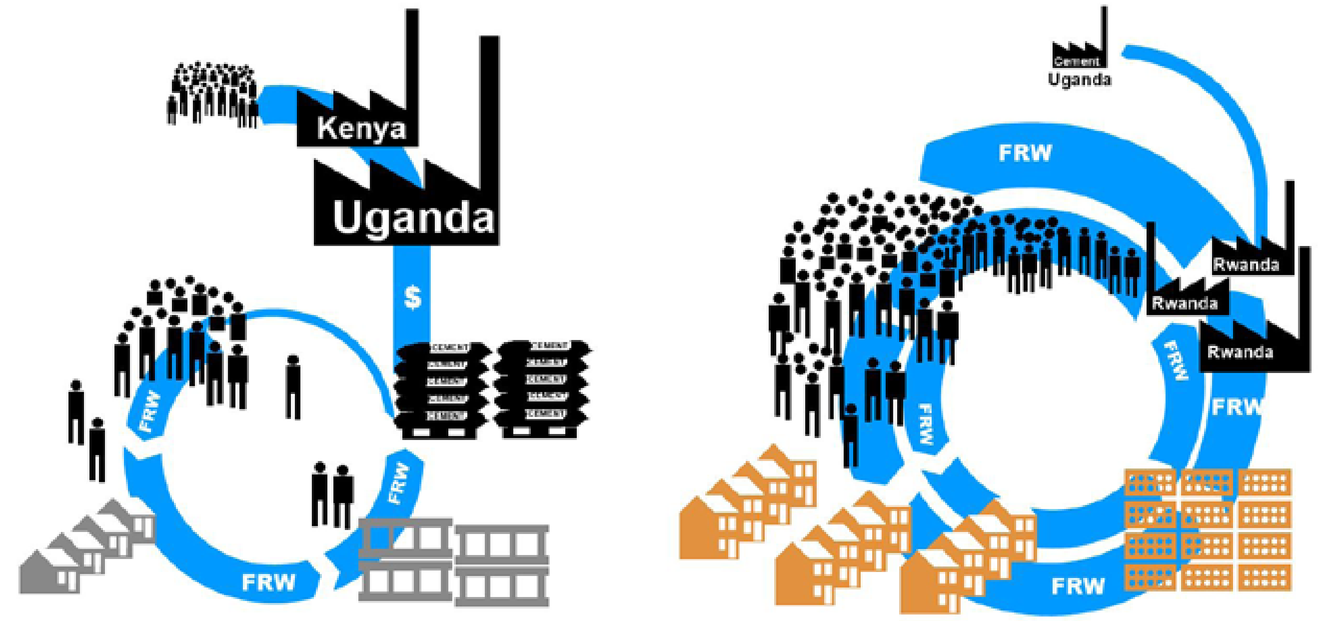
Even though the current production of Modern Bricks is still low, they can be - and, are - produced by existing small and medium-scale upgraded brickyards. 100 more could be upgraded and produce bricks for 3,000 houses per year. Given the annual demand of 40,000 - 50,000 new urban houses, **50** new medium-scale brick factories could profitably produce Modern Bricks to substitute the expensive and substandard traditional ones. In the case where houses currently made of imported cement would be built with Modern Bricks instead, up to **150** brick factories could operate profitably.



Brickyard Type	**XL** Semi-Industrial	**L** Artisanal	**M** Artisanal	**S** Upgraded Tile Kiln
Annual Output	3.000.000 Bricks (200-350 houses)	1.000.000 Bricks (70-110 houses)	600.000 Bricks (40-70 houses)	300.000 Bricks (20-35 houses)
New brickyards required for substituting:				
A: Traditional Bricks	50 brickyards - or-	150 brickyards - or -	190 brickyards - or -	380 brickyards
B: Cement Blocks	161 brickyards - or -	484 brickyards - or -	661 brickyards -or -	1,221 brickyards

REASON Nº 10: LOCAL BRICKS GENERATE LOCAL JOBS AND INCREASE THE MEDIUM INCOME

SUBSTITUTING FOREIGN CEMENT WITH MODERN BRICK COULD GENERATE UP TO 20,000 CONSTRUCTION JOBS



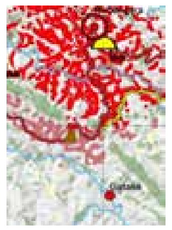
Current Situation

Concrete blocks are mostly composed of imported cement (Cimerwa production is sufficient for concrete work and mortar only). The capital spent for concrete blocks is mainly lost to the foreign cement industry and does not further circulate in the country or create local jobs.

Modern Bricks Replace Concrete Blocks

The introduction of **50** new modern brick facilities in proximity to Kigali and the secondary cities reduces the need for imported cement. With more production facilities in operation, the number of local jobs available in the building material and construction sector increases exponentially. The capital remains in-country, the job growth translates to higher demand.

The transformation of Rwanda's brick industry will require targeted interventions and investments all along the value chain. The Swiss Agency for Development Cooperation has therefore mandated Skat Consulting to offer the following services to entrepreneurs, investors and technicians in the following areas:

Site and Clay Analysis

Business Design Support


Access to Technologies + Skills

Marketing Support

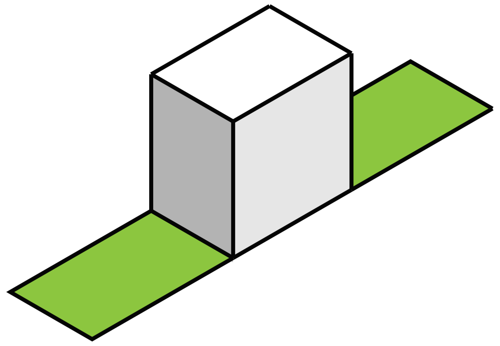
THE SWISS CUBE SYSTEM IS SUITABLE TO MOST PLOT AND TERRAIN CONFIGURATIONS

WHAT IS A SWISS CUBE?

The 'Swiss Cube' is a term used to described an optimised architecture, simple in shape, elegant in appearance and made of high quality materials.

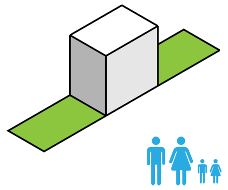
HOW DOES IT WORK?

The modular system is flexible and interchangeable, all while maintaining the same principle: 1 cube and 1 garden per family.

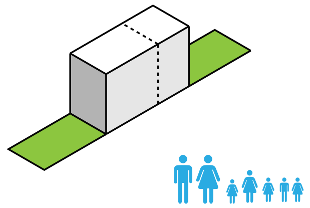


STRETCH

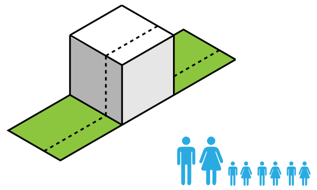
Extend or expand the cube to accommodate bigger families/budgets



S duplex



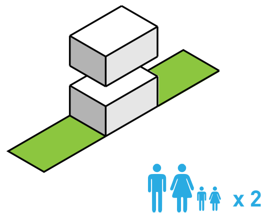
S Corner House



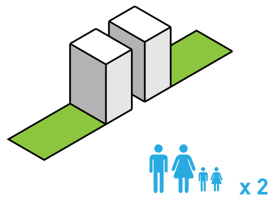
XXL duplex

SPLIT

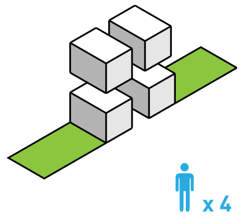
Maximize plot area by sudviding cube into smaller units



S superposed simplex



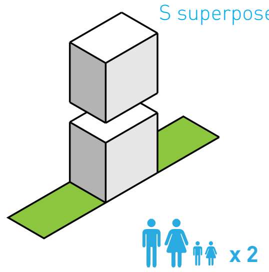
S back-to-back duplex



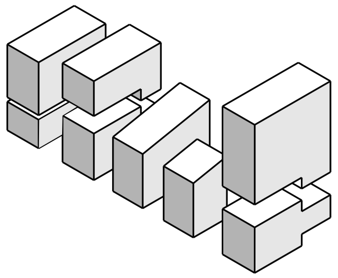
S studio

STACK

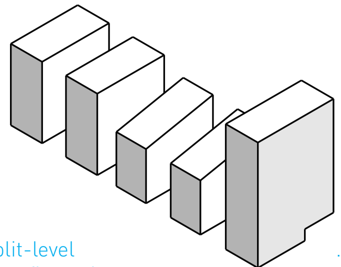
Superimpose the cube to maximize plot development without compromising on space minimizing

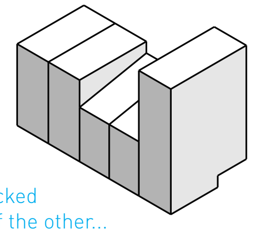


Stacked S duplexes



9 different split-level Swiss Cube configurations...





...can be stacked one on top of the other...

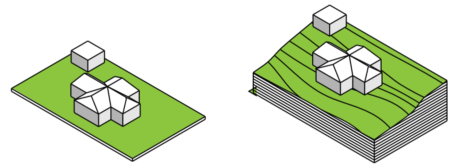
TAXONOMY
THE SWISS CUBE SYSTEM

THE SWISS CUBE SYSTEM IS SUITABLE TO MOST PLOT AND TERRAIN CONFIGURATIONS

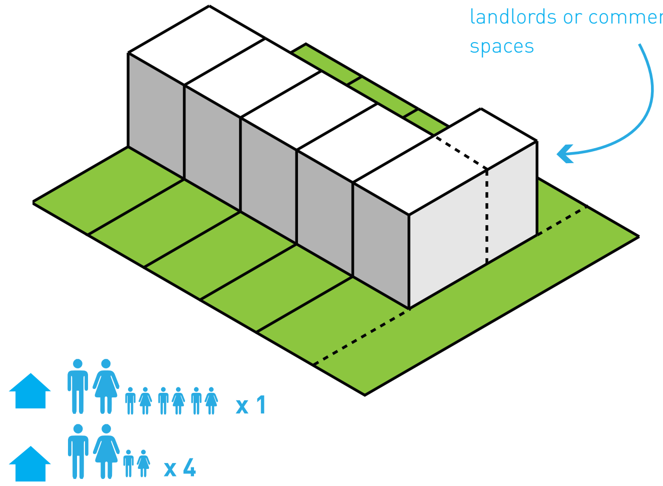
MULTIPLY

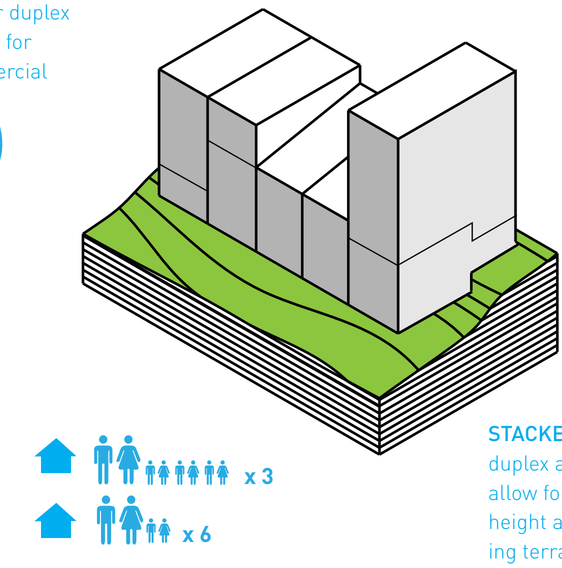
Maximize plot development and reduce construction costs by sharing partition walls and infrastructure services

Traditional plot developments are difficult to densify, especially on sloped terrains



STRETCHED corner duplex houses are suitable for landlords or commercial spaces

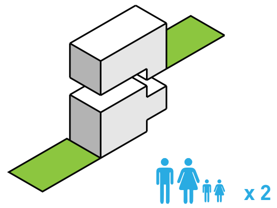




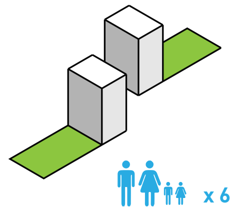
STACKED split-level simplex, duplex and triplex houses allow for increased density, height and luxury on challenging terrains.

SPLIT ON A SLOPE

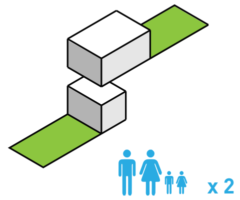
Maintain private garden access by introducing split-level units



S split-level simplexes



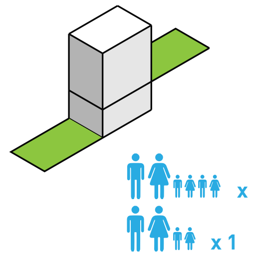
S Back-to-back split-level duplexes



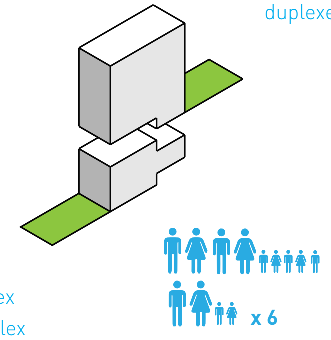
S studio
S simplex

STACK ON A SLOPE

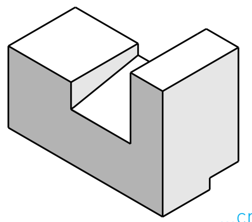
Increase density by superposing one unit on another



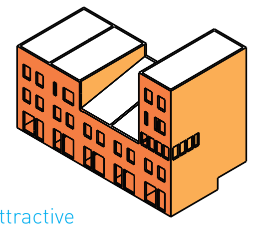
S stacked duplex
S stacked simplex



S stacked split-level triplex
S stacked split-level duplex



...and assembled into one volume...



...creating an attractive urban multiplex.

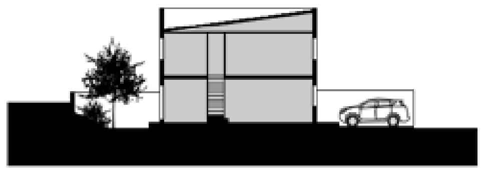
Anatomy of a building

Multiple units can be assembled on a standard plot without sacrificing basic amenities like direct road access and private garden

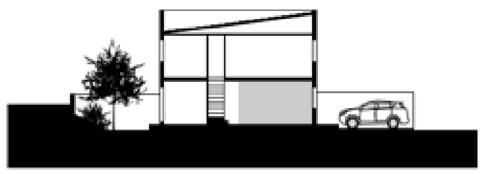
XXL

starting from
FRW **10.200.000**

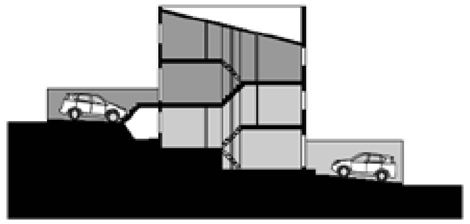
XXL SHELL	$86m^2$
Interior Dimensions: 5.20m x 8.34m	
Room Height: 2.40m	
Walling Material: Fully Facing Modern Bricks	
Slab: Maxspan	
Flooring: Cement Screed	
Roofing Material: Iron Sheet	



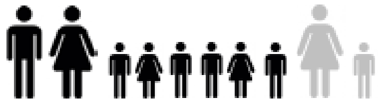
XXL Simple Duplex



XXL Apartment

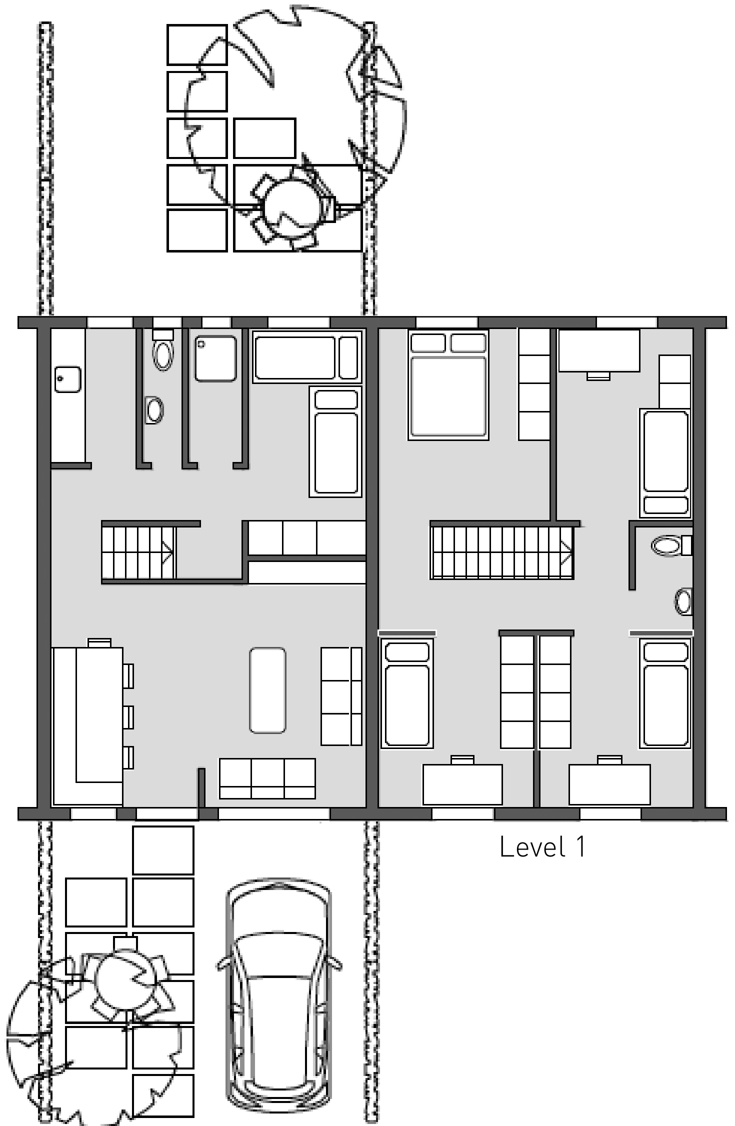


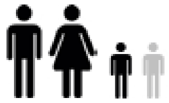
Double Split-level duplex



starting from
FRW **14.400.000**

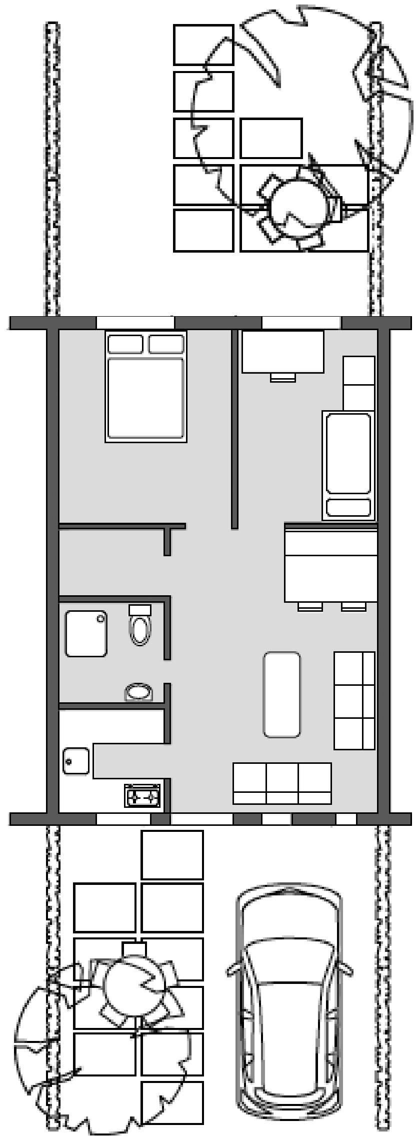
XXL DUPLEX	$86m^2$
Living Room/Dining Room	20
Master Bedroom	10
Additional Bedrooms (x4)	32
Kitchen	√
Bathroom	√
Storage	√
Garden	√

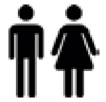




starting from
FRW **8.600.000**

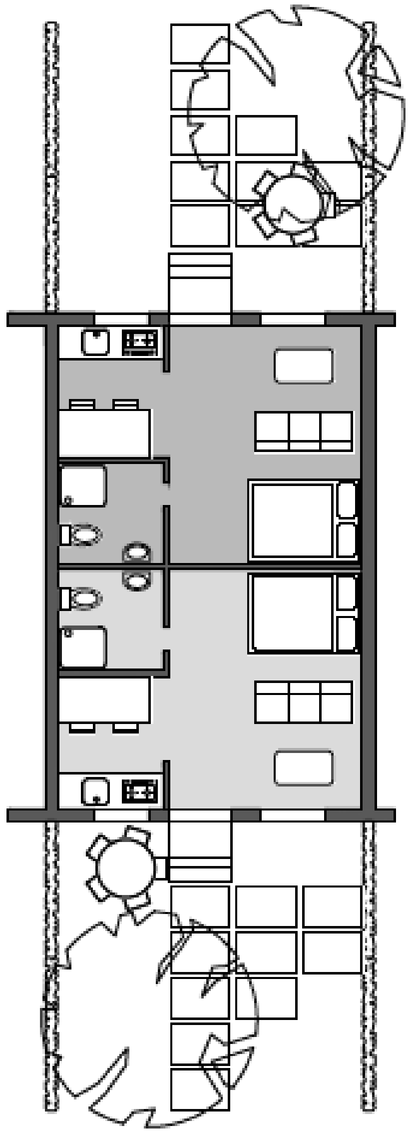
XXL SIMPLEX (1 unit per floor)	$43m^2$
Living Room/Dining Room	17
Master Bedroom	10
Additional Bedrooms (x1)	8
Kitchen	√
Bathroom	√
Storage	√
Garden	√





starting from
FRW **4.200.000**

XXL STUDIO (2 units per floor)	$21m^2$
Living Room/Dining	13
Kitchen	√
Bathroom	√
Garden	√



XXL

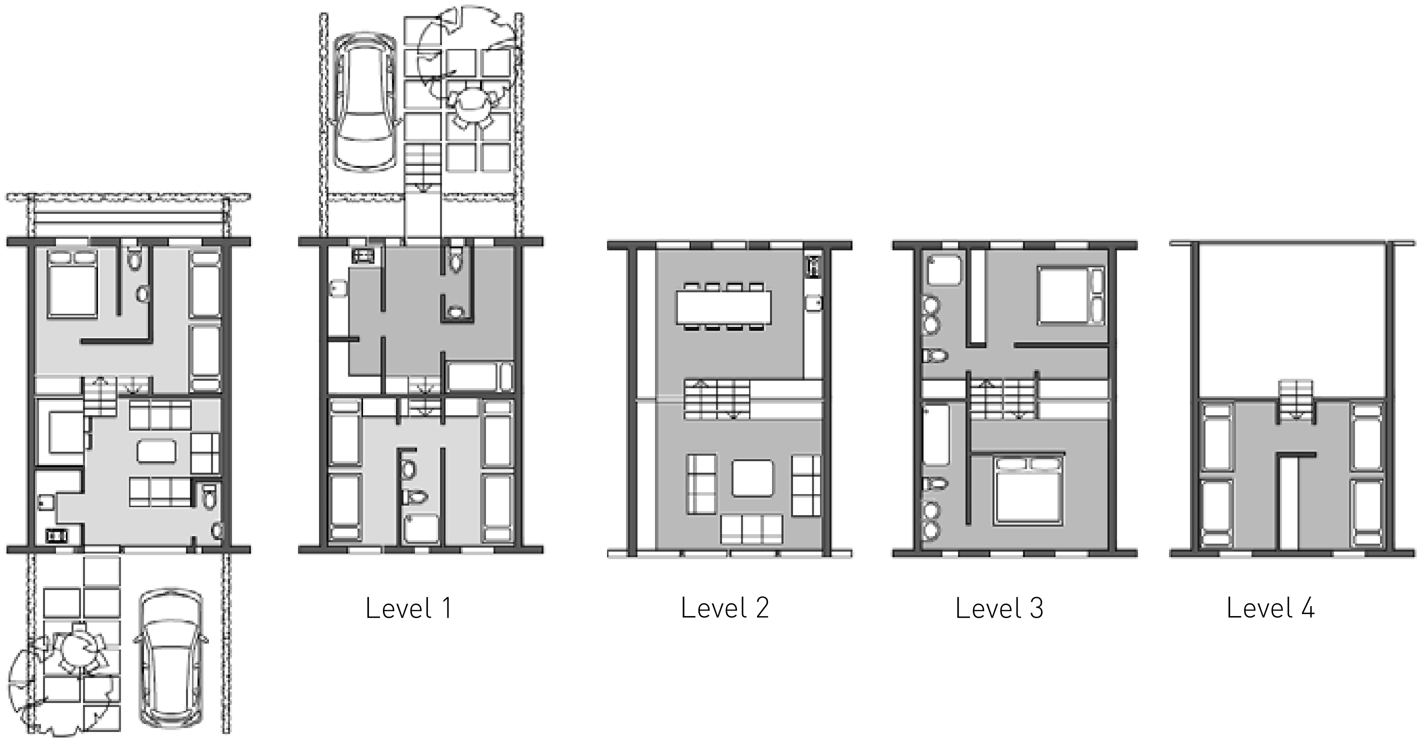
x 8

starting from
FRW 12.000.000

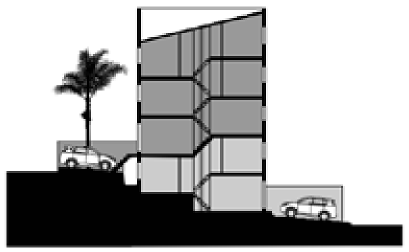
XXL 8-in-1 DUPLEX	64m^2
Living Room / Dining Room	16
Master Bedroom	8
Additional Bedrooms (x3)	24
Kitchen	√
Bathroom (x3)	√
Storage	√
Garden	√

starting from
FRW 25.000.000

XXL 8-in-1 TRIPLEX	130m^2
Living Room/Dining Room	43
Master Bedroom	13
Additional Bedrooms (x4)	35
Kitchen / Storage	8
Bathroom (x3)	√
Storage	√
Garden	√



Unit 1 Level 0 - 1
Unit 2 Level 1 - 4



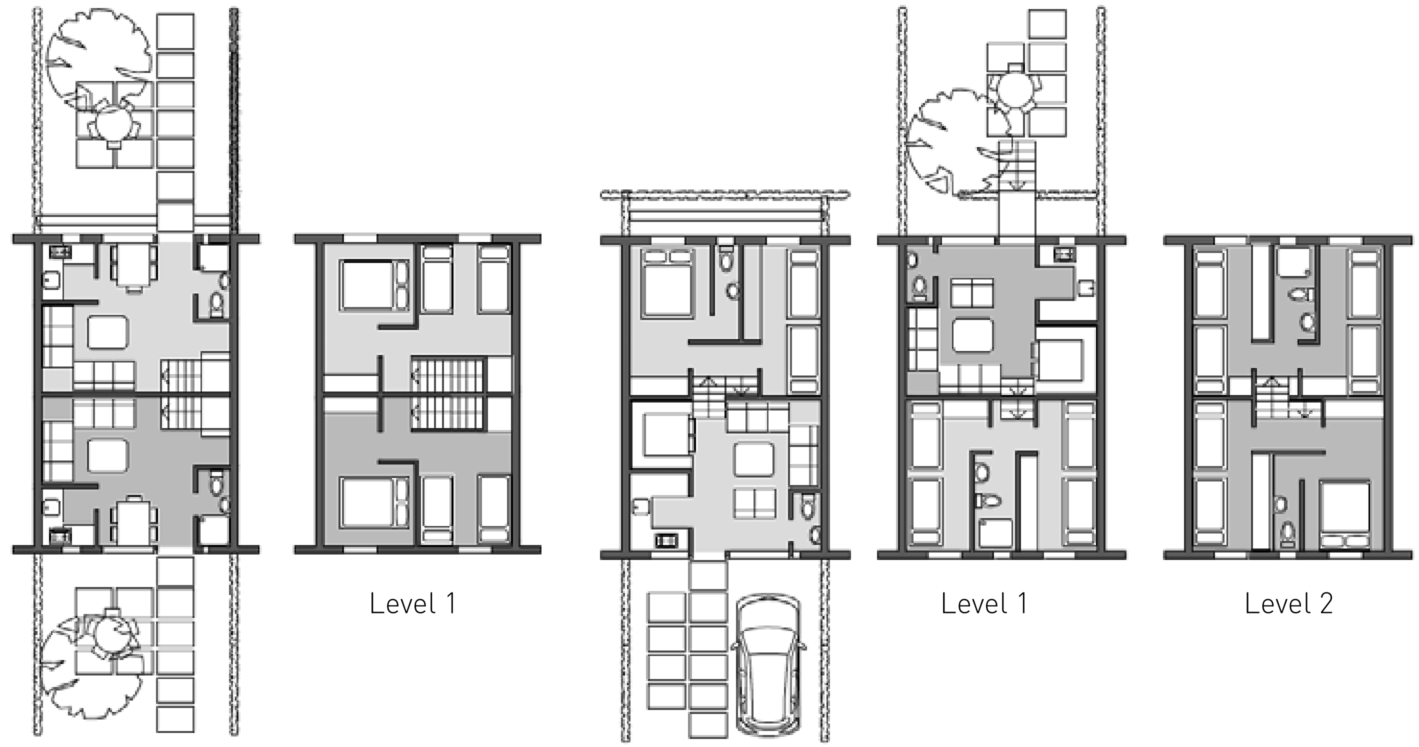
XXL Split-level Duplex and Triplex

starting from
FRW 8.000.000 / unit

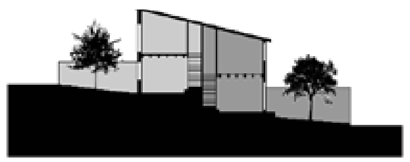
XXL BACK-TO-BACK DUPLEX (1 of 2 units)	43m^2
Living Room/Dining Room	17
Master Bedroom	8
Additional Bedrooms (x1)	8
Kitchen	√
Bathroom	√
Storage	√
Garden	√

starting from
FRW 12.000.000 / unit

XXL DOUBLE SPLIT LEVEL DUPLEX (1 of 2 units)	64m^2
Living Room/Dining Room	16
Master Bedroom	8
Additional Bedrooms (x3)	24
Kitchen	√
Bathroom (x3)	√
Storage	√
Garden	√

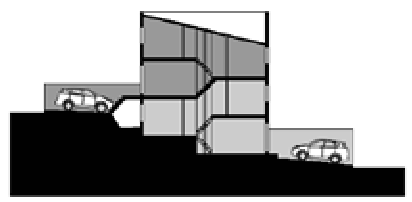


Unit 1
Unit 2



XXL Split-level Back-to-Back

Unit 1 L0 - L1
Unit 2 L1 - L2



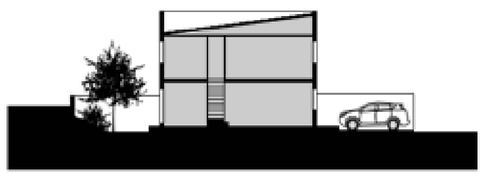
XXL Double Split-level Duplex

XL

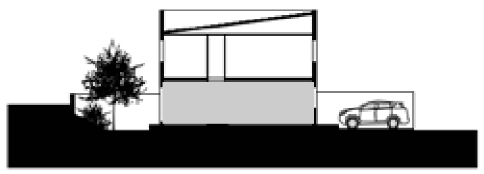
starting from
FRW **9.800.000**

XL SHELL	81m²

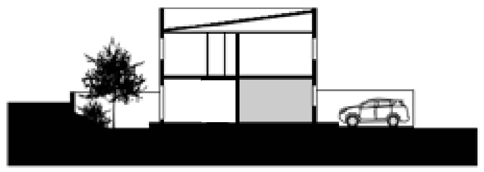
Interior Dimensions: 4.92m x 8.34m
Room Height: 2.40m
Walling Material: Fully Facing Modern Bricks
Slab: Maxspan
Flooring: Cement Screed
Roofing Material: Iron Sheet



XL Duplex



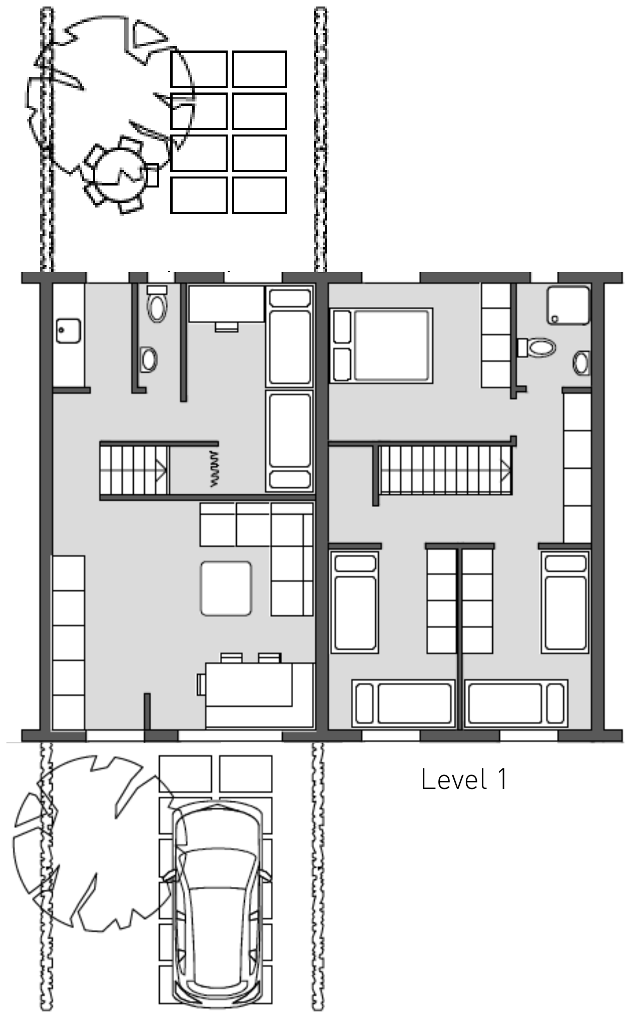
XL Simplex



XL Studio

starting from
FRW **13.600.000**

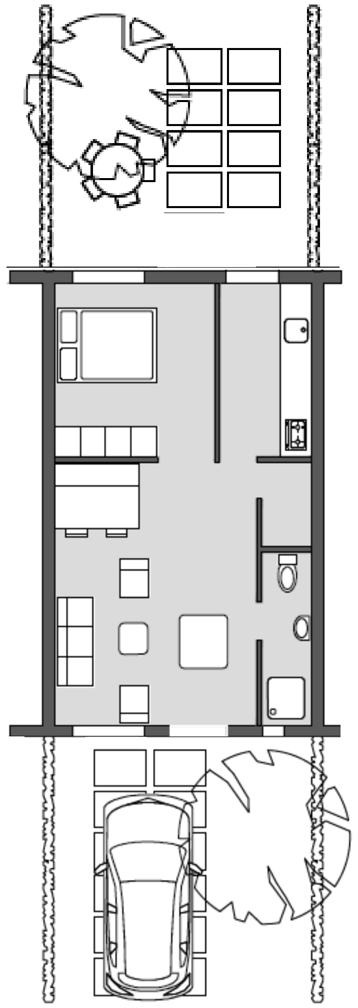
XL DUPLEX	81m²
Living Room/Dining Room	21
Master Bedroom	10
Additional Bedrooms (x3)	24
Kitchen	√
Bathroom	√
Storage	√
Garden	√



Level 0

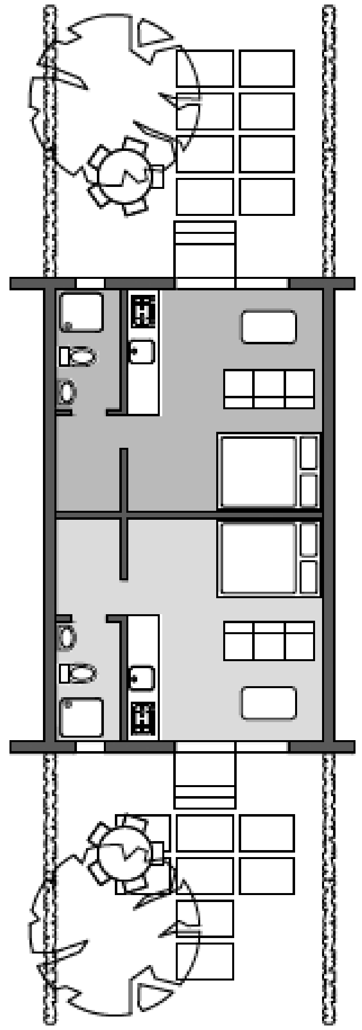
starting from
FRW **8.000.000**

XL SIMPLEX	40m²
Living Room/Dining Room	15
Master Bedroom	10
Kitchen	√
Bathroom	√
Storage	√
Garden	√



starting from
FRW **4.000.000**

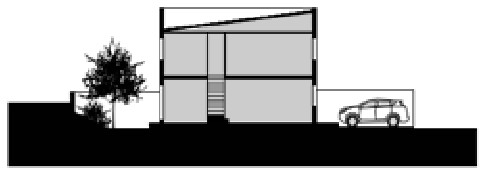
XL STUDIO (1 of 2 units)	20m²
Living Room/Bedroom	14
Kitchen	√
Bathroom	√
Garden	√



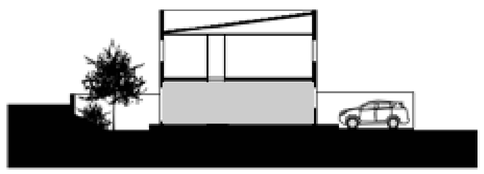
L

starting from
FRW **9.500.000**

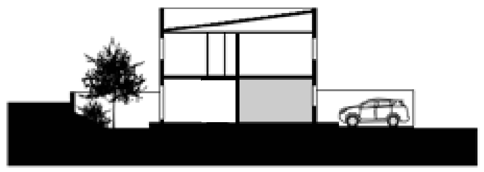
L SHELL	77m²
Interior Dimensions: 4.63m x 8.34m	
Room Height: 2.40m	
Walling Material: Fully Facing Modern Bricks	
Slab: Maxspan	
Flooring: Cement Screed	
Roofing Material: Iron Sheet	



L Single Duplex



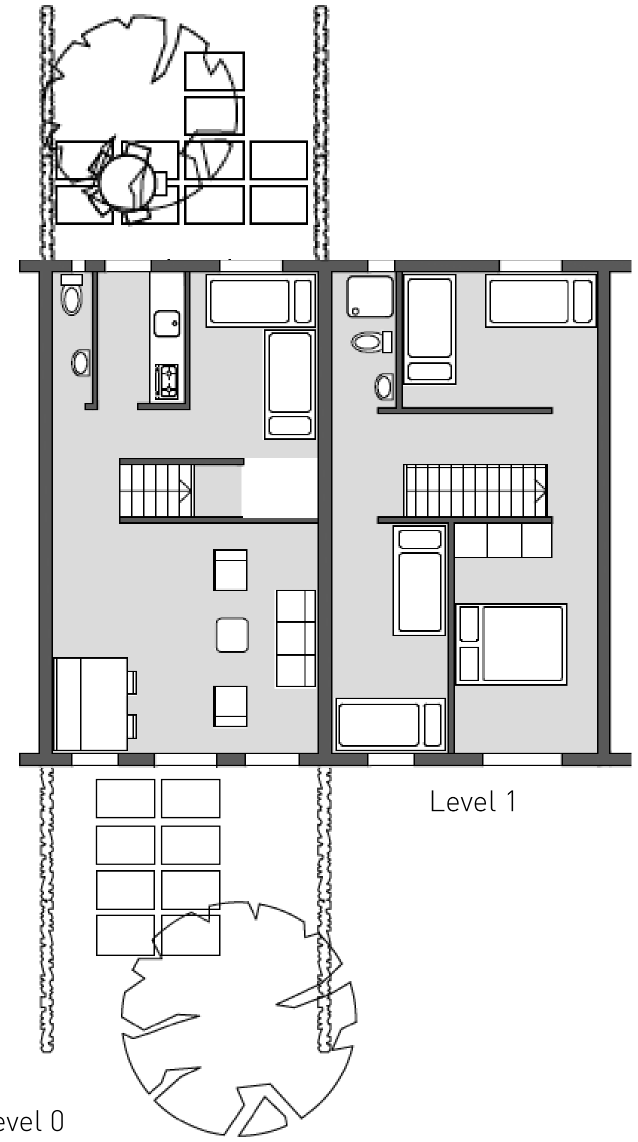
L Simplex



L Studio

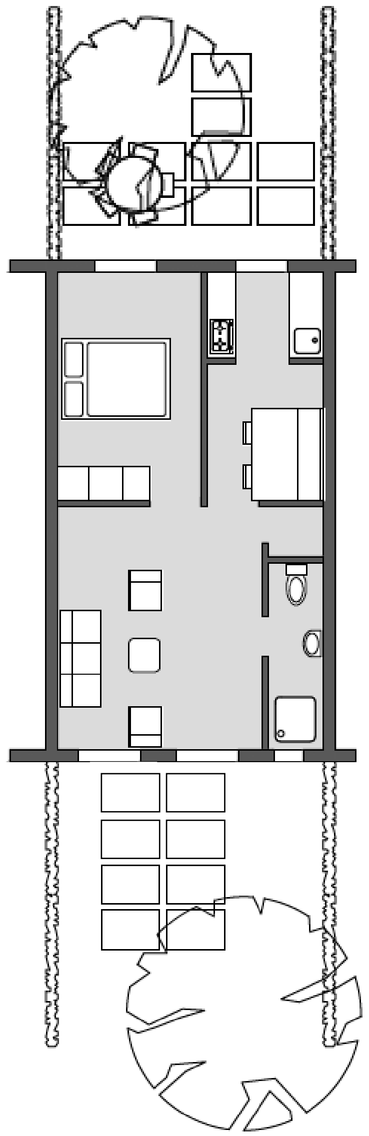
starting from
FRW **13.000.000**

L DUPLEX	77m²
Living Room/Dining Room	20
Master Bedroom	10
Additional Bedrooms (x3)	24
Kitchen	√
Bathroom	√
Storage	√
Garden	√



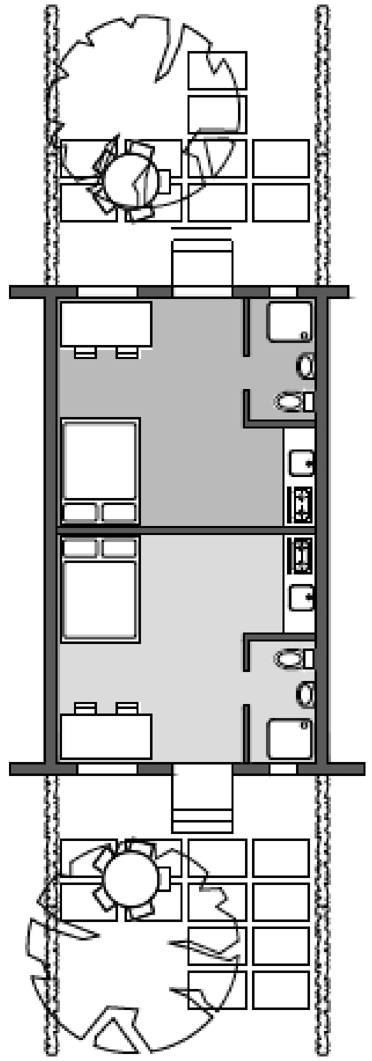
starting from
FRW **7.500.000**

L SIMPLEX	38m²
Living Room/Dining Room	15
Master Bedroom	10
Additional Bedrooms	8
Kitchen	√
Bathroom	√
Storage	√
Garden	√



starting from
FRW **3.800.000 / unit**

L STUDIO (1 of 2 units)	19m²
Living Room/Bedroom	14
Kitchen	√
Bathroom	√
Garden	√

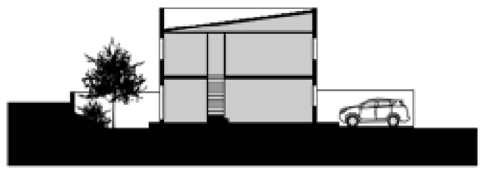


M

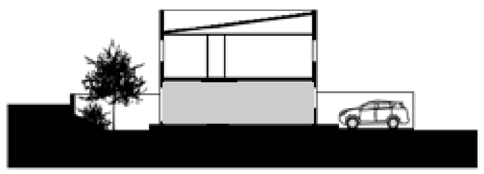
starting from
FRW **8.100.000**

M SHELL	$67m^2$

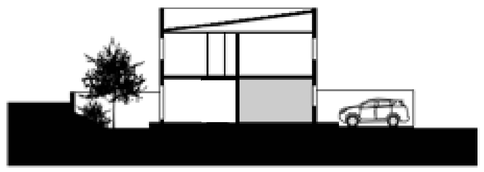
Interior Dimensions: 4.06m x 8.34m
Room Height: 2.40m
Walling Material: Fully Facing Modern Bricks
Slab: Maxspan or Timber Floor
Flooring: Cement Screed
Roofing Material: Iron Sheet



XS Simple Duplex



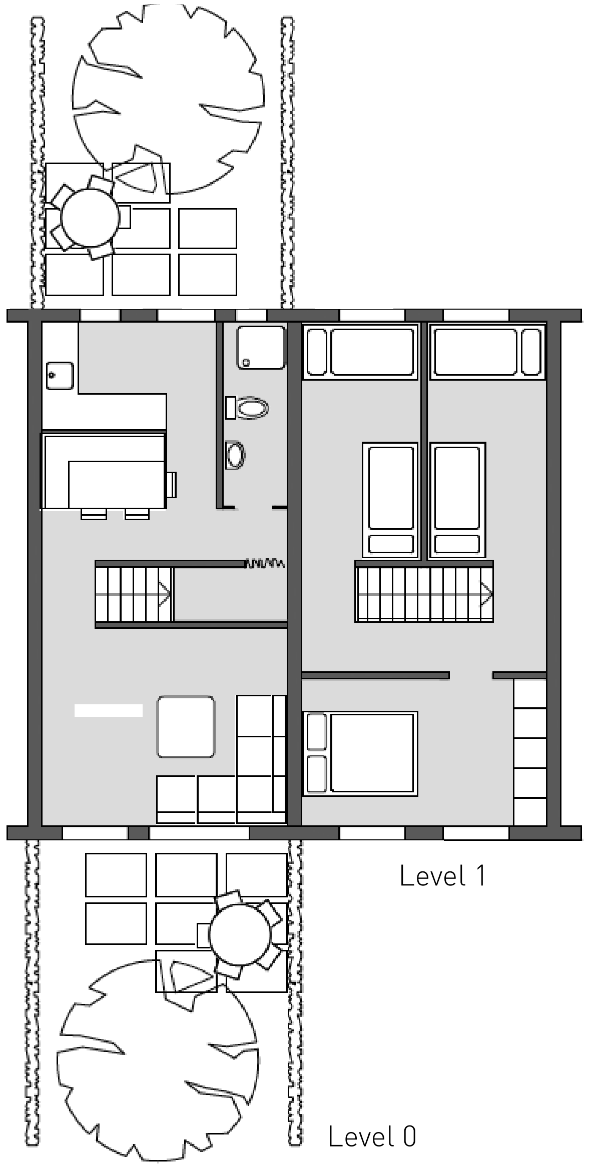
M Simplex



M Studio

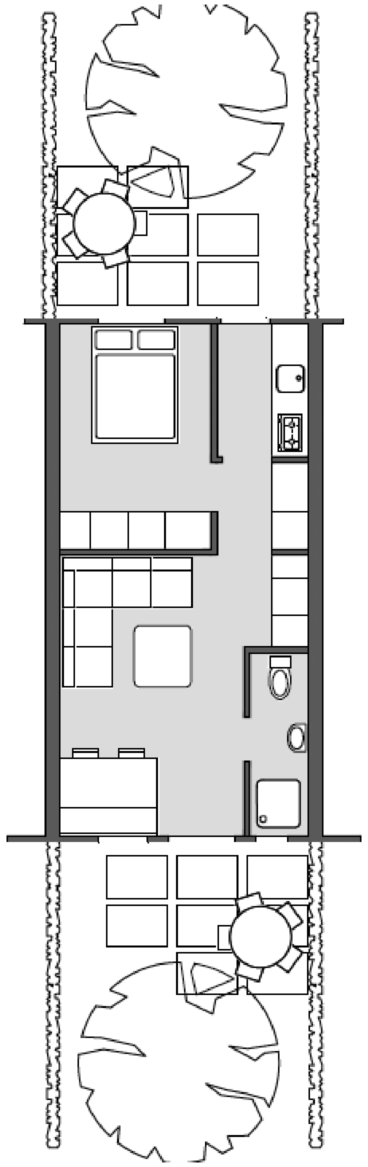
starting from
FRW **10.700.000**

M DUPLEX	$67m^2$
Living Room/Dining Room	20
Master Bedroom	10
Additional Bedrooms (x2)	16
Kitchen	√
Bathroom	√
Storage	√
Garden	√



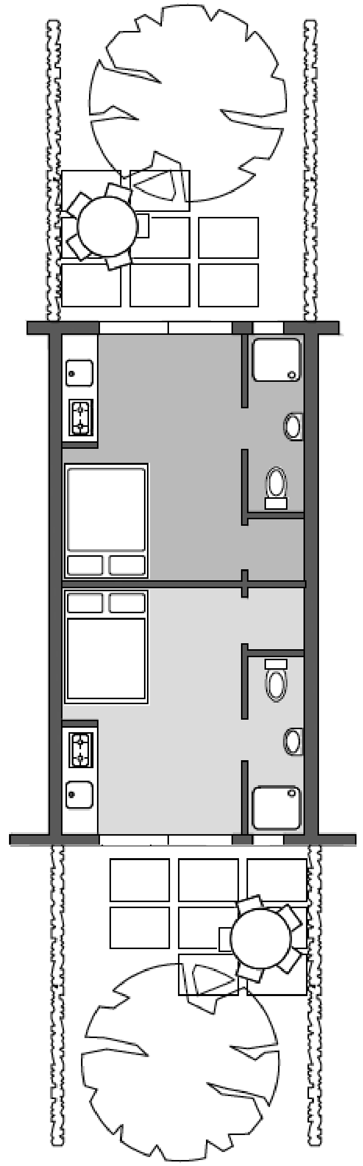
starting from
FRW **6.500.000**

M SIMPLEX	$33m^2$
Living Room/Dining Room	15
Master Bedroom	9
Kitchen	√
Bathroom	√
Storage	√
Garden	√



starting from
FRW **3.600.000 / unit**

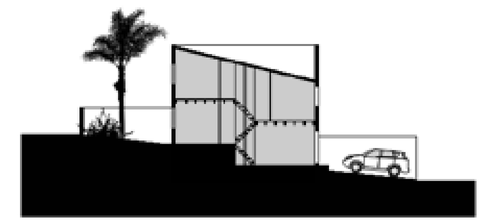
M STUDIO (1 of 2 units)	$16m^2$
Living Room/Bedroom	11
Kitchen	√
Bathroom	√
Garden	√



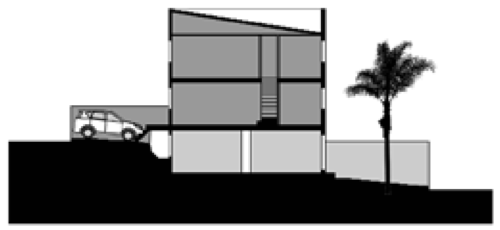
S

starting from
FRW **7.500.000**

S SHELL	$63m^2$
Interior Dimensions: 3.78m x 8.34m	
Room Height: 2.40m	
Walling Material: Fully Facing Modern Bricks	
Slab: Timber Floor / Maxpan between units	
Flooring: Cement Screed	
Roofing Material: Iron Sheet	



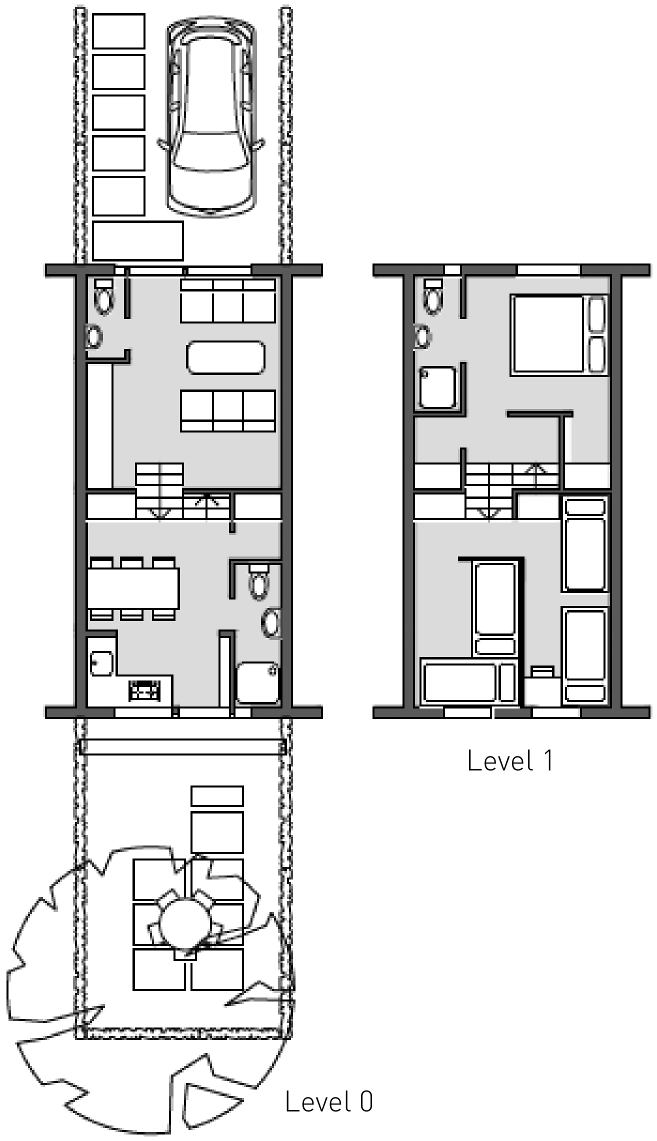
S Single Split-level Duplex



S Simplex + S Duplex

starting from
FRW **10.500.000**

S SINGLE SPLIT-LEVEL DUPLEX	$63m^2$
Living Room / Dining Room	20
Master Bedroom	9 -11
Additional Bedrooms (x2)	16 (or 7+7)
Kitchen	√
Bathroom (x2 or x3)	√
Storage	√
Garden	√

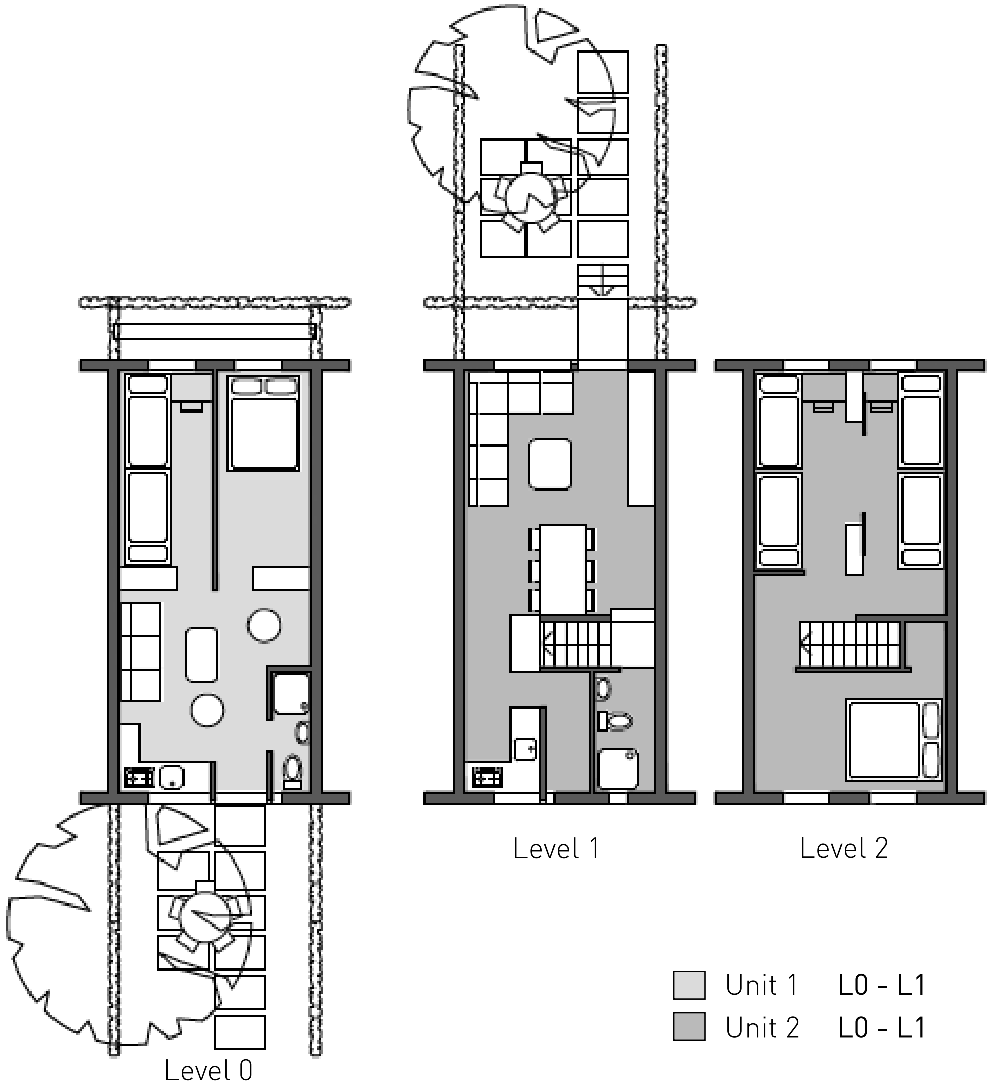


starting from
FRW **5.000.000**

S SIMPLEX	$31m^2$
Living Room / Dining Room	13
Master Bedroom	8
Additional Bedroom (x1)	8
Kitchen	√
Bathroom	√
Storage	√
Garden	√

starting from
FRW **9.500.000**

S DUPLEX	$63m^2$
Living Room / Dining Room	19
Master Bedroom	10
Additional Bedrooms (x2)	16
Kitchen	√
Bathroom (x1)	√
Garden	√



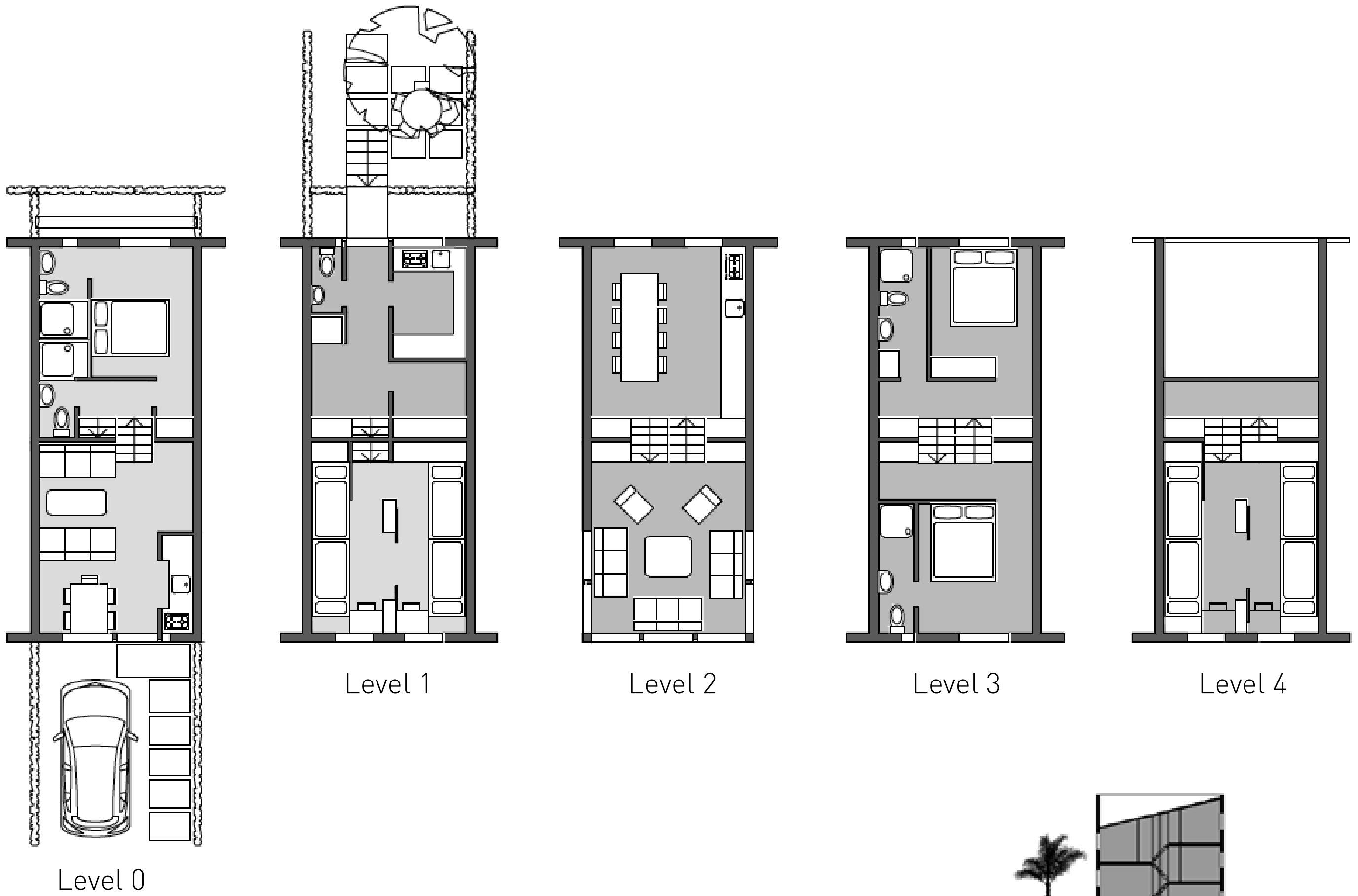
S

starting from
FRW **9.000.000**

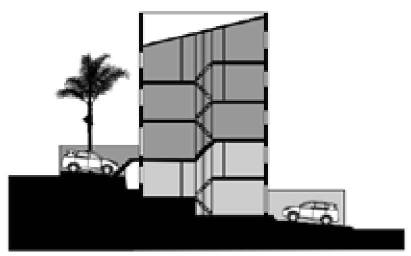
S SLOPE SPLIT LEVEL DUPLEX (Unit 1)	$54m^2$
Living Room/Dining Room	13.5
Master Bedroom	9
Additional Bedrooms (x2)	15 (or 6 +7)
Kitchen	√
Bathroom (x2)	√
Storage	√
Garden	√

starting from
FRW **18.000.000**

S SLOPE SPLIT LEVEL TRIPLEX (Unit 2)	$107m^2$
Living Room / Dining Room	36
Master Bedroom	10
Additional Bedrooms (x3)	24
Kitchen / Storage	8
Bathroom (x3)	√
Garden	√



Unit 1 L0 - L1
Unit 2 L1 - L4



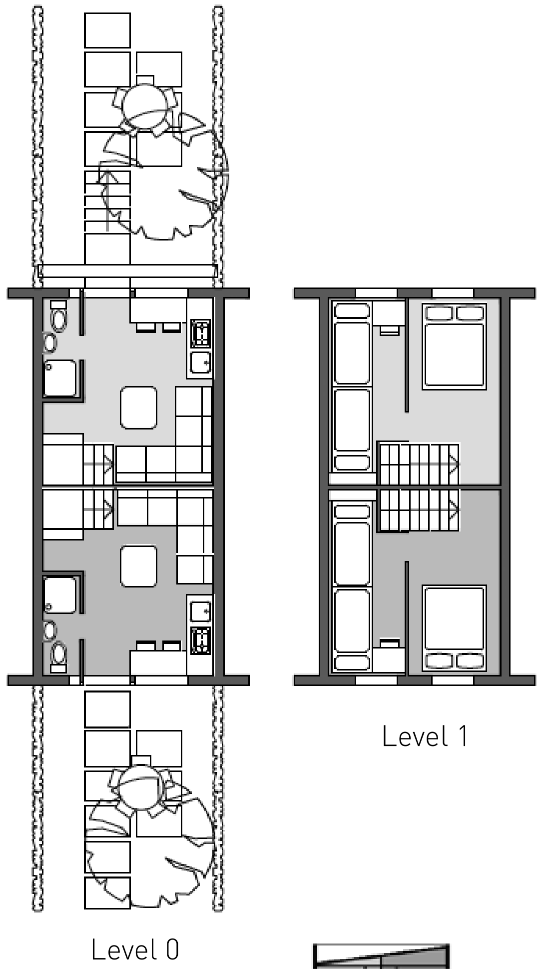
S Split-level Duplex and Triplex

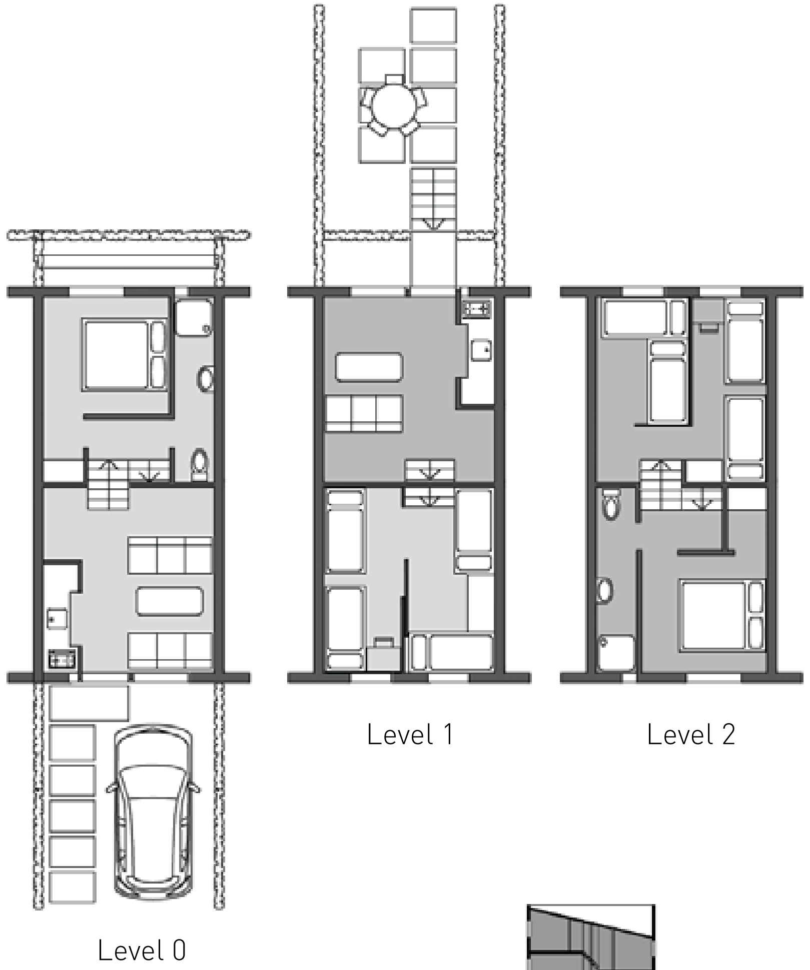
starting from
FRW **6.000.000 / unit**

S BACK-TO-BACK DUPLEX (1 of 2 units)	$31m^2$
Living Room/Dining Room	13
Bedroom (1 or 2)	12
Kitchen	√
Bathroom (x1)	√
Garden	√

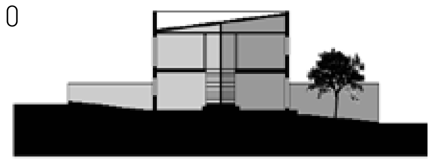
starting from
FRW **9.000.000 / unit**

S DOUBLE SPLIT-LEVEL DUPLEX (1 of 2 units)	$54m^2$
Living Room	16
Master Bedroom	9
Additional Bedrooms (x2)	16
Kitchen	√
Bathroom (x1)	√
Garden	√



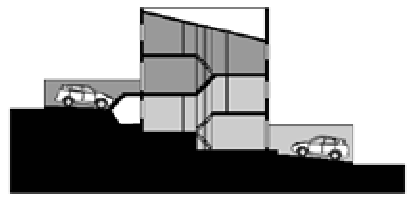


Unit 1
Unit 2



S Back-to-Back

Unit 1 L0 - L1
Unit 2 L1 - L2



S Double Split-level Duplex

XS

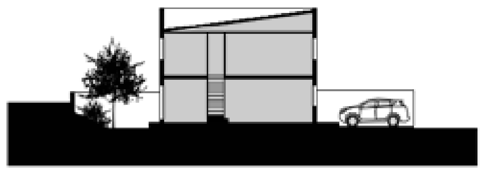
starting from
FRW **7.500.000**

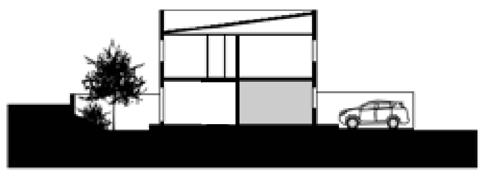
XS SHELL	58m²

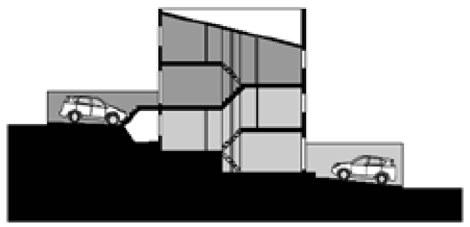
Interior Dimensions: 3.50m x 8.34m
Room Height: 2.40m
Walling Material: Fully Facing Modern Bricks
Slab: Timber Floor / Maxpan between stacked units
Ground Floor Slab: Cement Screed
Roofing Material: Iron Sheet

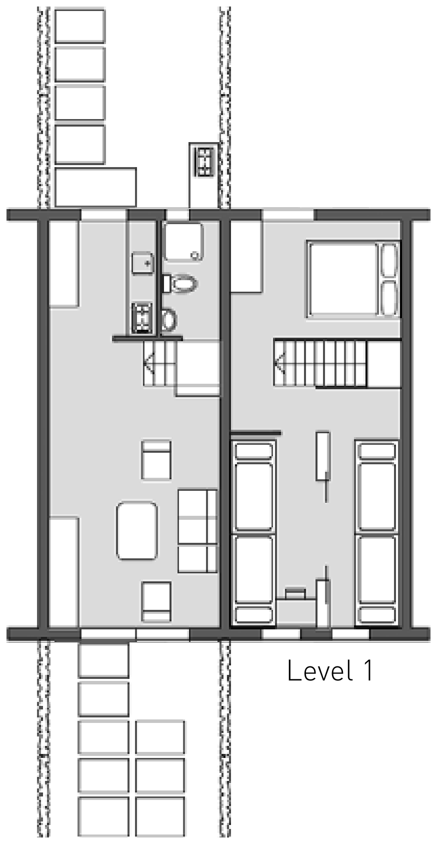
starting from
FRW **9.200.000**

XS SIMPLE DUPLEX	58m²
Living Room/Dining Room	19
Master Bedroom	8
Additional Bedrooms (2x)	15
Kitchen	√
Bathroom (1x)	√
Storage	√
Garden	√


XS Simple Duplex


XS Studio


XS Double Split-level duplex

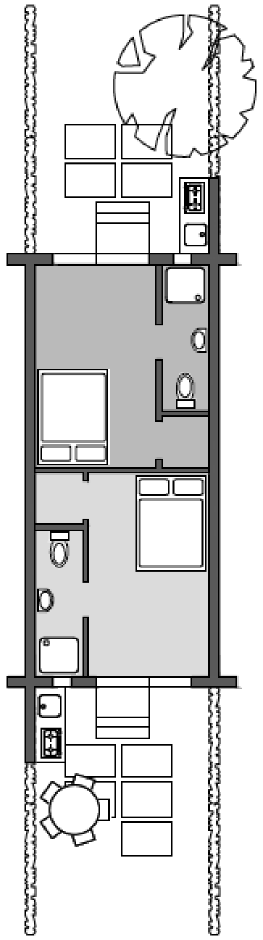


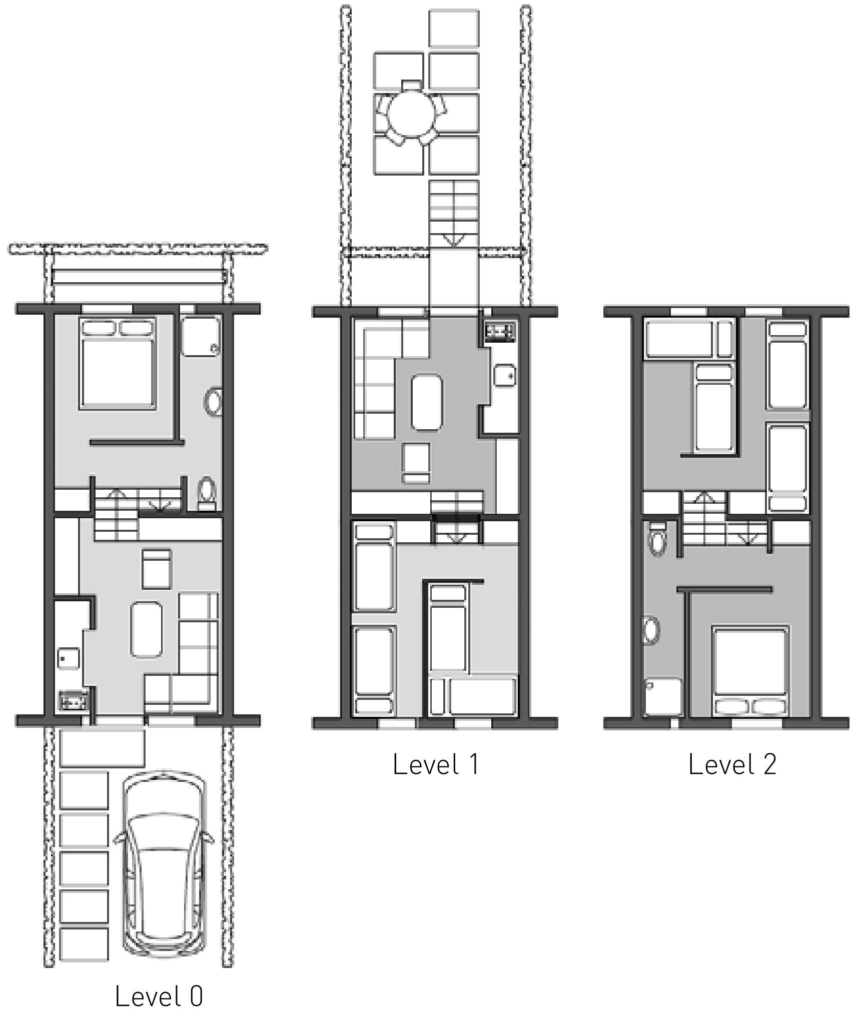
starting from
FRW **2.900.000 / unit**

XS STUDIO (1 of 2 units)	14m²
Living Room/Dining Room	11
Kitchen	√
Bathroom	√
Garden	√

starting from
FRW **8.000.000 / unit**

XS DOUBLE SPLIT-LEVEL DUPLEX (1 of 2 units)	43m²
Living Room/Bedroom	12.5
Master Bedroom	8
Additional Bedrooms (x2)	15
Kitchen	√
Bathroom (x1)	√
Garden	√



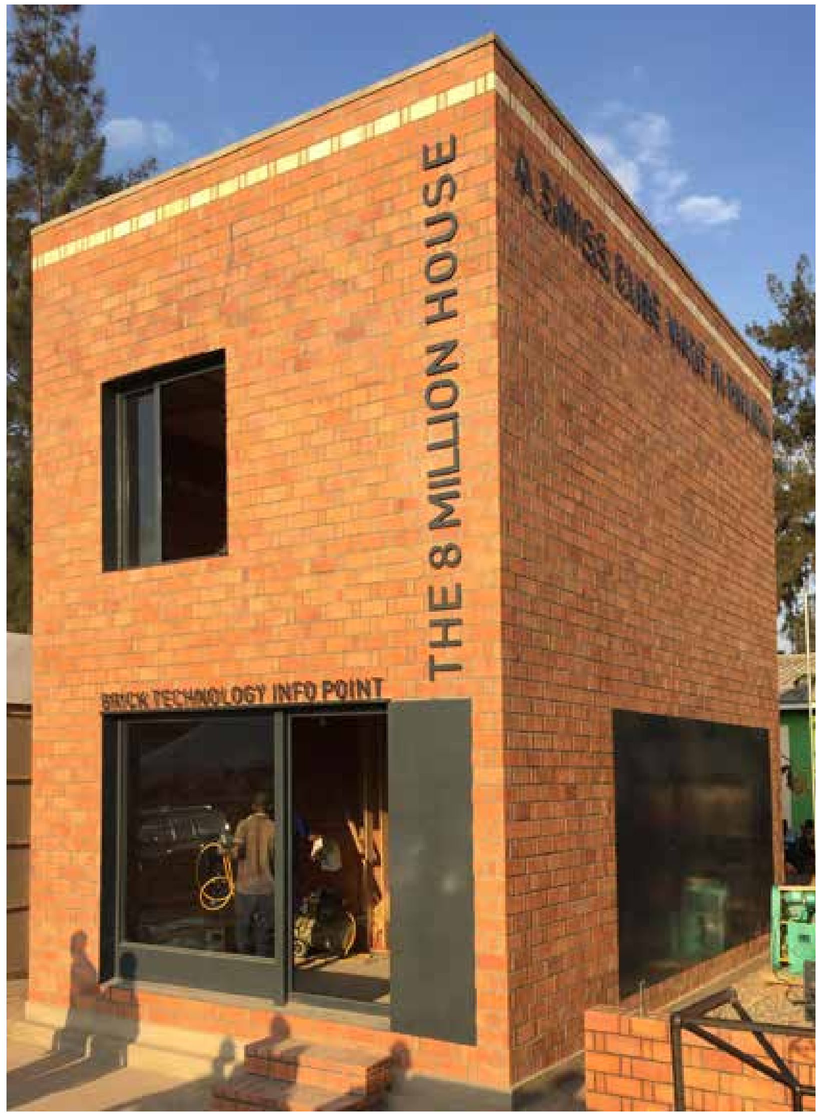


Unit 1 L0 - L1
Unit 2 L1 - L2

THE "SWISS CUBE" DEMONSTRATES THE POTENTIAL OF THE LOCAL INDUSTRY TO SUPPLY AFFORDABLE HOUSING

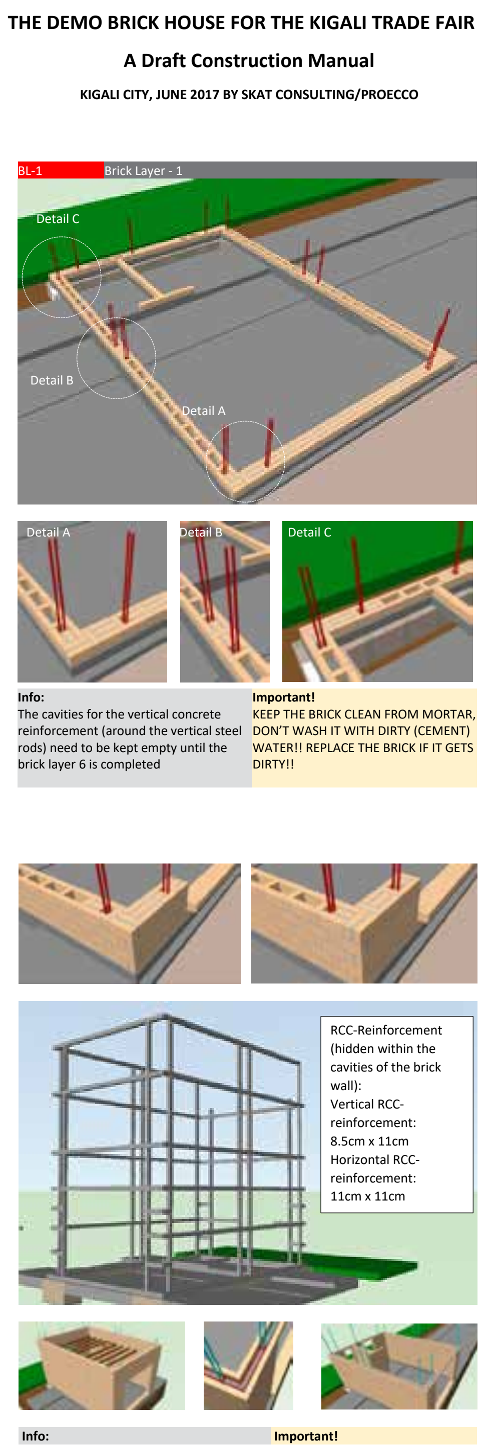
Area: 50 m^2
Unit Cost: FRW 8 million (basic finishes)
Cost includes all features except land and engineering. Profit and labor are included.
Cost per square meter: 190 USD

Elements Tested:
RCC Reinforced Rowlock Bond Wall
Timber Slab + Timber Stairs
Innovation:
Illustrated Construction Guide (layer by layer details)



Elevation of a 5-in-1 Rental Building on a 15m x 25m plot: 280m^2 / FRW 45 mio

The Owners's Corner house or Shop 80-95m^2: FRW 13-15mio | Rental Unit 1 50m^2/8 mio | Rental Unit 2 50m^2/8 mio | Rental Unit 3 50m^2/8 mio | Rental Unit 4 50m^2/8 mio



CASE STUDY
KIGALI PSF EXPO HOUSE (2017)

DETAILED CONSTRUCTION MANUAL GUIDES EXECUTION WORKS AND SUPERVISION

The Kigali PSF Demonstration House introduces the "S.M.A.R.T. Tafali Etage" concept to the general public. Made entirely of local materials, it displays the potential of what the Rwandan construction industry would be capable of supplying on mass scale, if all relevant stakeholders, from the brickmaker to engineers and architects, worked together to address the high demand for quality, affordable construction for rapidly urbanizaing districts, cities and towns.

RUSIZI MODEL BRICK DUPLEX SHOPHOUSE (2015)

TESTING COST SAVINGS POTENTIAL AND PRACTICABILITY OF ROWLOCK BOND CONSTRUCTION

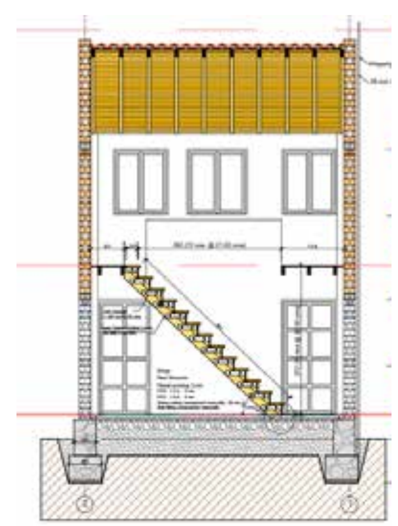
Built in 2015, the Rusizi Modern Brick Duplex Shophouse serves as a testing/display unit for cost-effective housing solutions built of Modern Bricks. It also serves as a model for a mixed-use building for urban contexts.

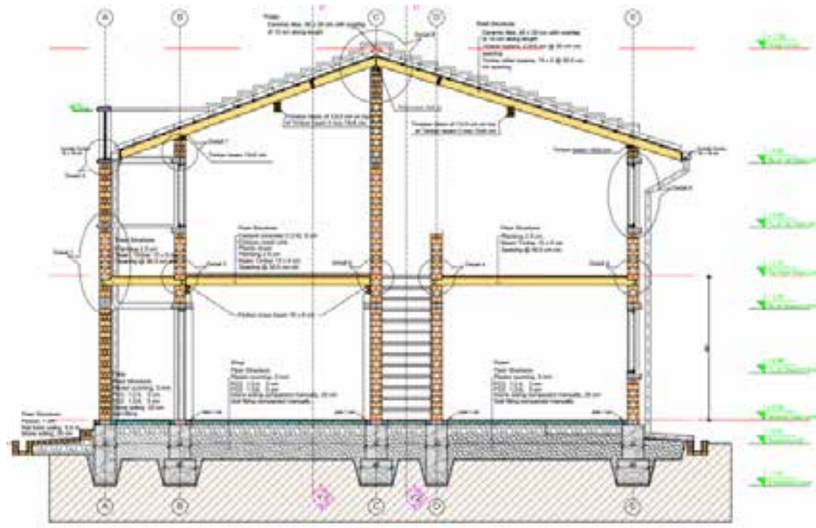
RUSIZI MODEL BRICK DUPLEX SHOPHOUSE (2015)

SHOPHOUSE SERVES AS EXAMPLE OF ANCHOR BUILDING FOR ROWHOUSE TYPOLOGY

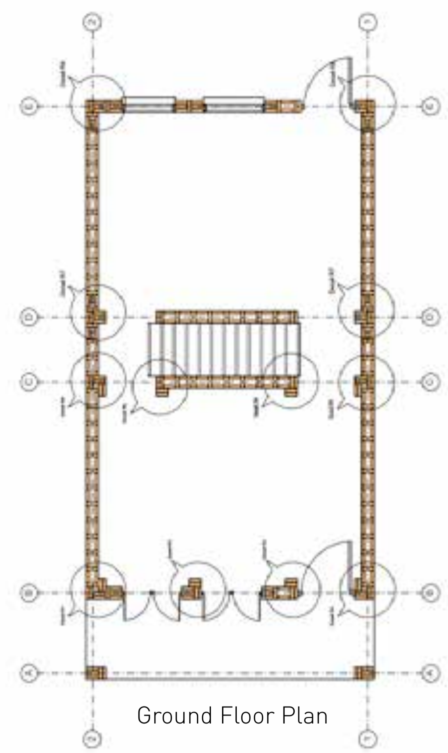
Area: 117 m^2
Unit Cost: FRW 20 million (basic finishes)
Cost includes all features except land and engineering. Profit and labor are included.
Cost per square meter: 206 USD

Elements Tested:
RCC Reinforced Rowlock Bond Wall
Timber Slab with Terracotta + Plastic Finish
Timber Stairs
Strawtec Partitioning Walls





Tranverse Cross Section (above)
Longitudinal Cross Section (right)



Ground Floor Plan

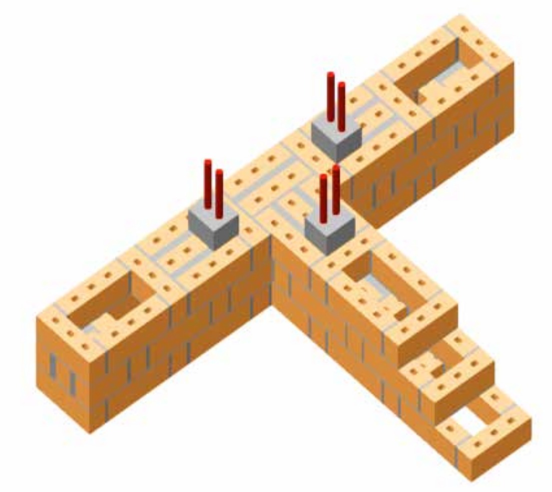
ANNEX 1

RCC REINFORCED ROWLOCK BOND WALLING SYSTEM

ROWLOCK BOND: A TIME-TESTED CONSTRUCTION METHOD

Rowlock Bond walling is a cost-effective walling system for houses up to 2.5 stories. It was popular during the industrial revolution in both the UK and United States. In the last three decades, the system has made a resurgence in South Asia. A damage assessment after the Kathmandu Earthquake (Nepal 2015) proved the strength and good para-seismic performance of the Rowlock Bond walling system. The system has now been officially endorsed by the Nepalese government.


Canada, 1858

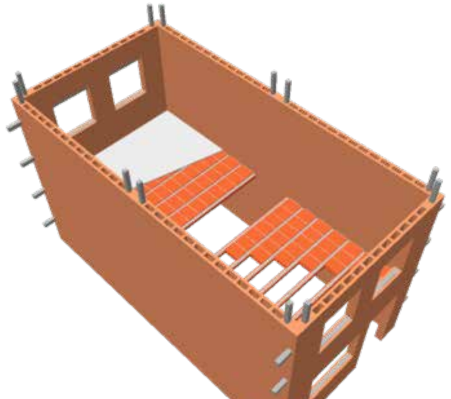

Joint Detail of Modern Semi-Industrial
RCC Reinforced Rowlock Bond Brick Wall

ANNEX 2

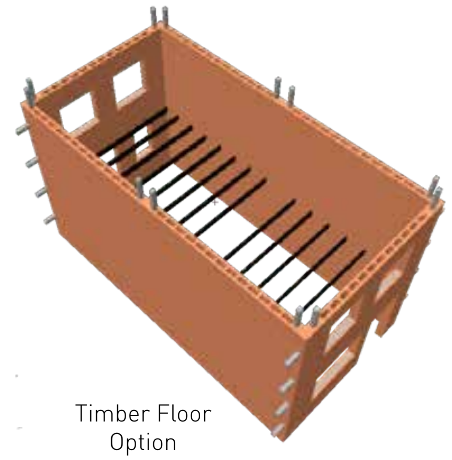
MODERN BRICK MULTIPLEX CONSTRUCTION SYSTEM

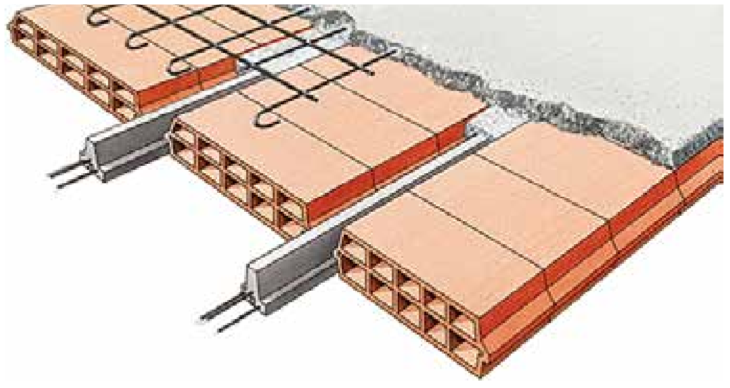
STRUCTURAL INTEGRITY ALLOWS FOR MAXIMUM FLEXIBILITY

The Modern Brick Muliplex Construction System is a "strong box" held together by concrete reinforcement (tie beams). The result is a structural frame within which flooring and walling elements can be adjusted and modified at will. All typologies are suitable for maxpan floor slabs, while the narrower models, M and S, can be outfitted with a timber floor. Both systems can be applied without modification to the structural "box."

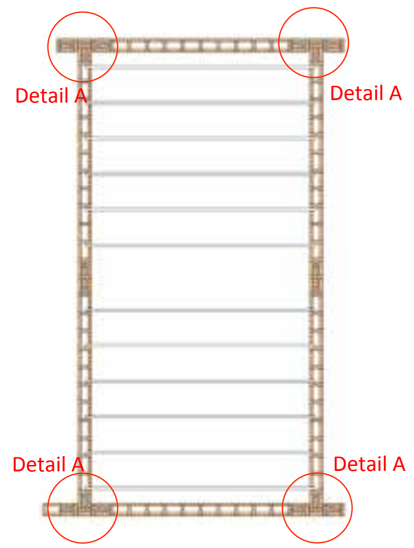

Maxpan Slab
Option

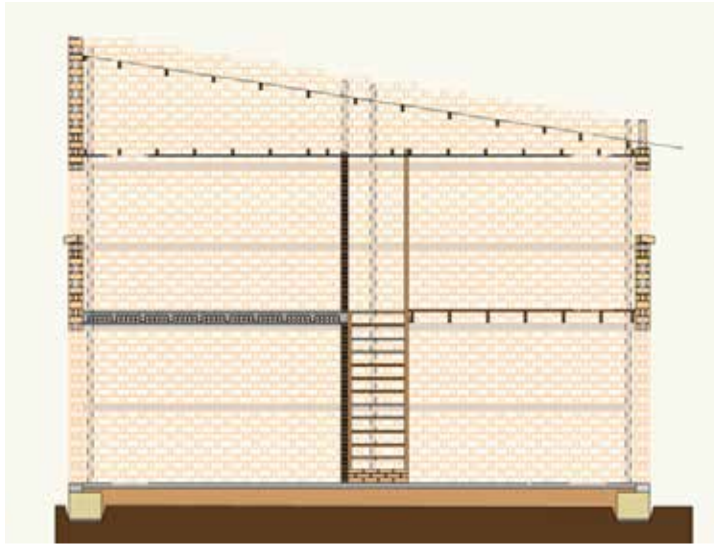

Hidden RCC
Reinforcement


Timber Floor
Option


Maxpan Slabs are suitable for all unit sizes (XXL through S).


Timber Floors suitable for unit sizes M and S.





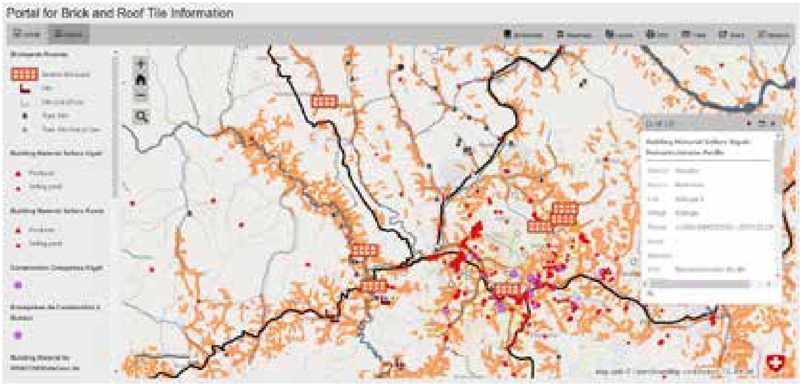
ANNEX 3

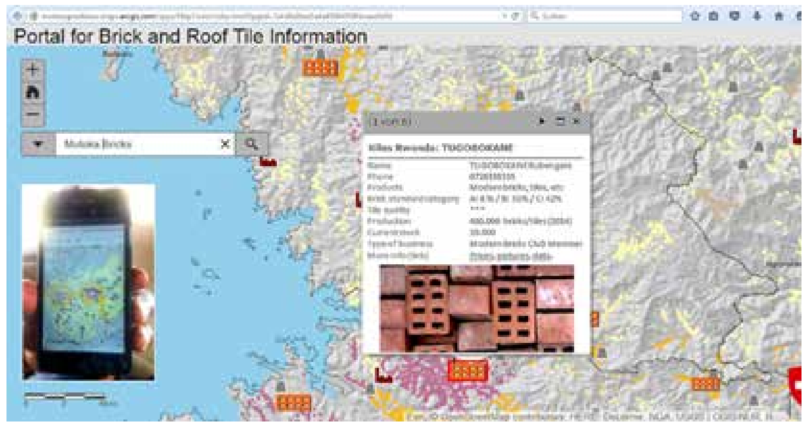
CONSTRUCTION INDUSTRY INFO PORTAL

PLATFORM FEATURES CONSTRUCTION INDUSTRY INFORMATION FOR AUDIENCES IN THE GREAT LAKES REGION

The Service Portal "Made in Great Lakes" offers access to information on various sectors and products in the Great Lakes Region relevant for building material producers, contractors, developers and authorities. Key features include real-time data on brick supply, downloadable demand and supply scenario projection tools and a list of regional construction industry events.

madeingreatlakes.com





KEY PORTAL FEATURES:

Document Library featuring urban planning codes, regulations and laws

Tenders for construction and infrastructure projects in the Secondary Cities and Districts

Training Manuals on construction material production and application

National Maps of Zones Suitable for Semi-Industrial Brick Production in the Great Lakes

Location services allowing construction firms to locate modern brick makers next to their sites

Production Zones of Building Material Producers and Resellers displayed on a map of clay areas suitable for semi-industrial production.

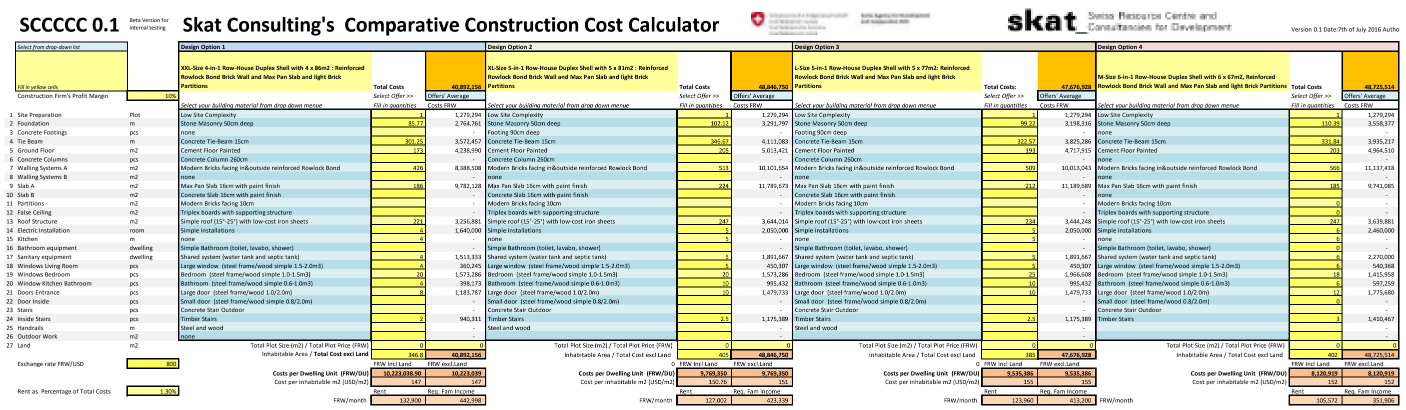
Address Book of construction industry stakeholders

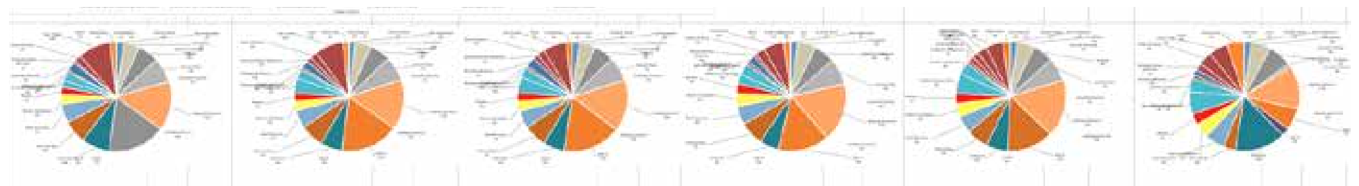
Statistics on regional building material production quantities;

Downloadable tools used to simulate building material demand and to cost compare of construction costs

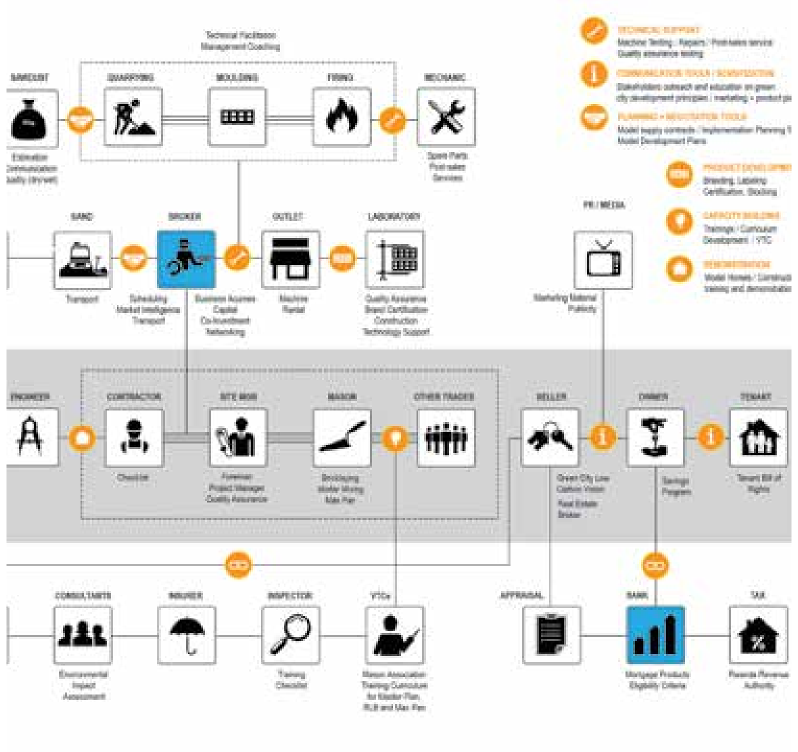
PORTAL INCLUDES REAL-TIME BRICK SUPPLY DATA, STATISTICS AND ESTIMATION TOOLS

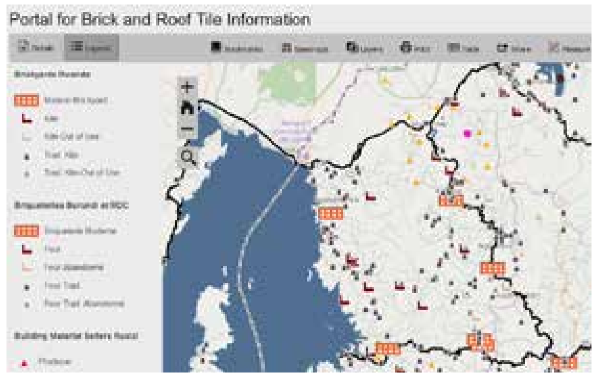
The Portal displays maps and data collected and analysed by the programme, in particular potential sites for clayish soils extraction, and suitable fuels. One of the newest features is the Comparative Construction Cost Calculator, which allows technicians and clients to compare relative costs of different construction technologies and building materials for any given project.





In addition to the Excel-based tools, the WebPortal features access to information about affordable housing construction value chains, training manuals for designers, planners and builders, along with a link to Buildapedia, the PROECCO project's online database of building material producers and construction techniques.

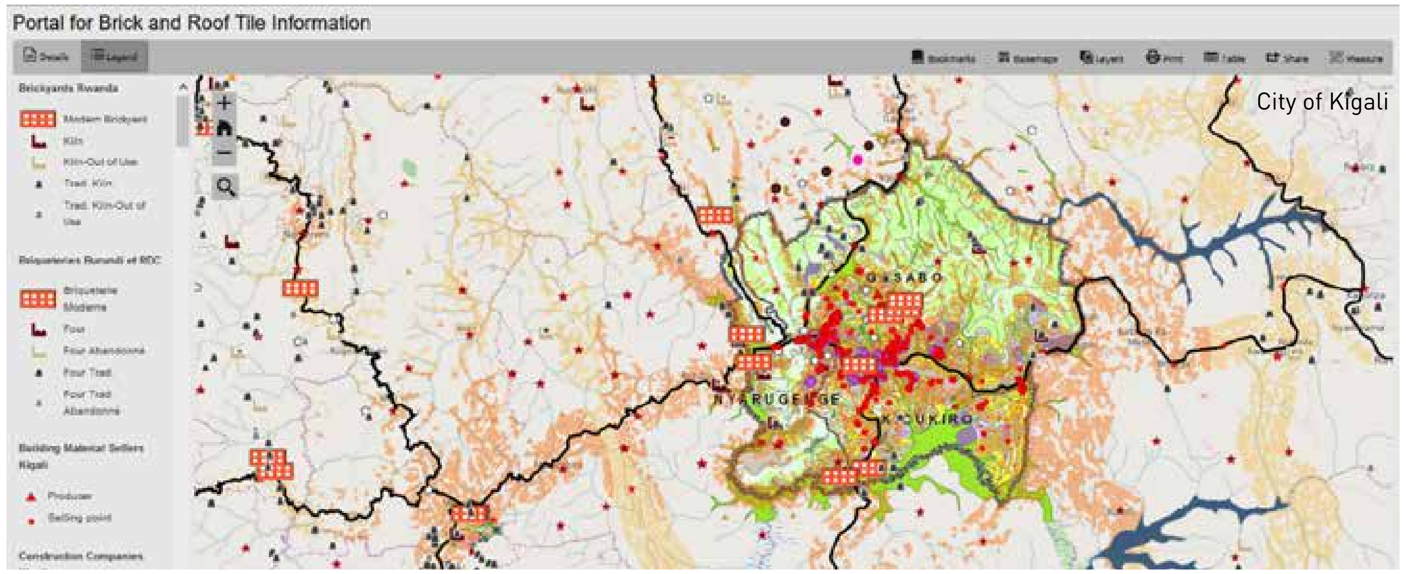


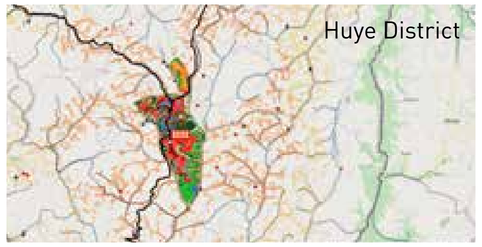


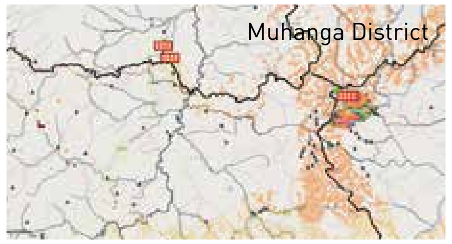
ANNEX 4 BASE MAPS FOR DEVELOPING MODERN BRICK SUPPLY FOR URBAN AGGLOMERATIONS

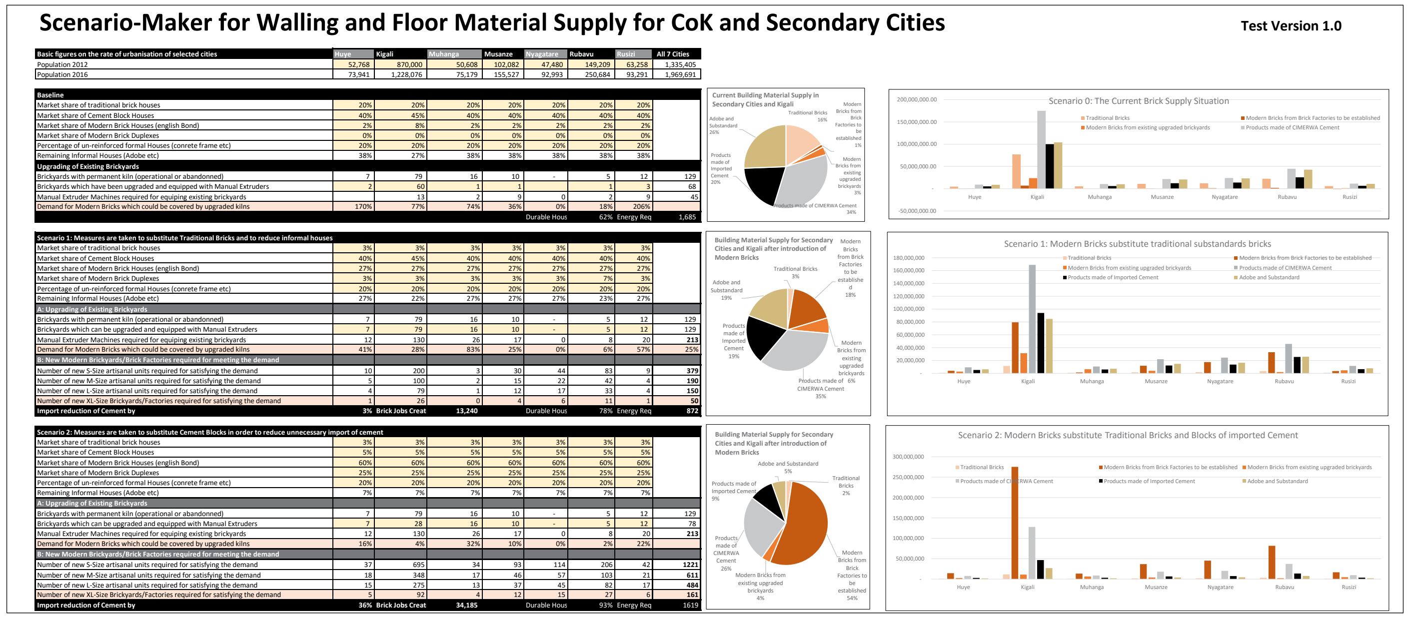
BRICK SECTOR DATA OVERLAID WITH MASTER PLAN LAND USE SPECIFICATIONS

Since 2012, the PROECCO project has consistently collected data and mapped location information for producers of brick and tiles across Rwanda. This information is overlaid on land use plan to facilitate the urbanization agenda. It follows that in 2016, the project introduced the Scenario Maker Tool, a tool that estimates the future demand of building materials in City of Kigali and the 6 Secondary Cities.



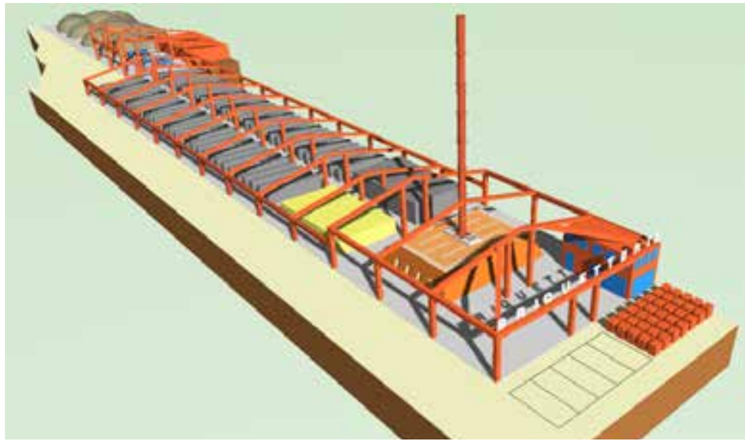


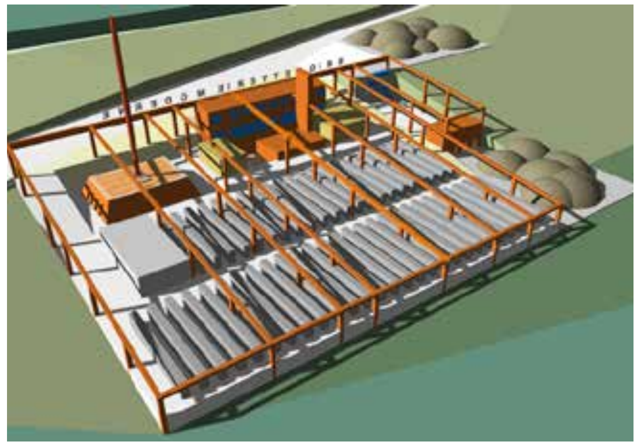


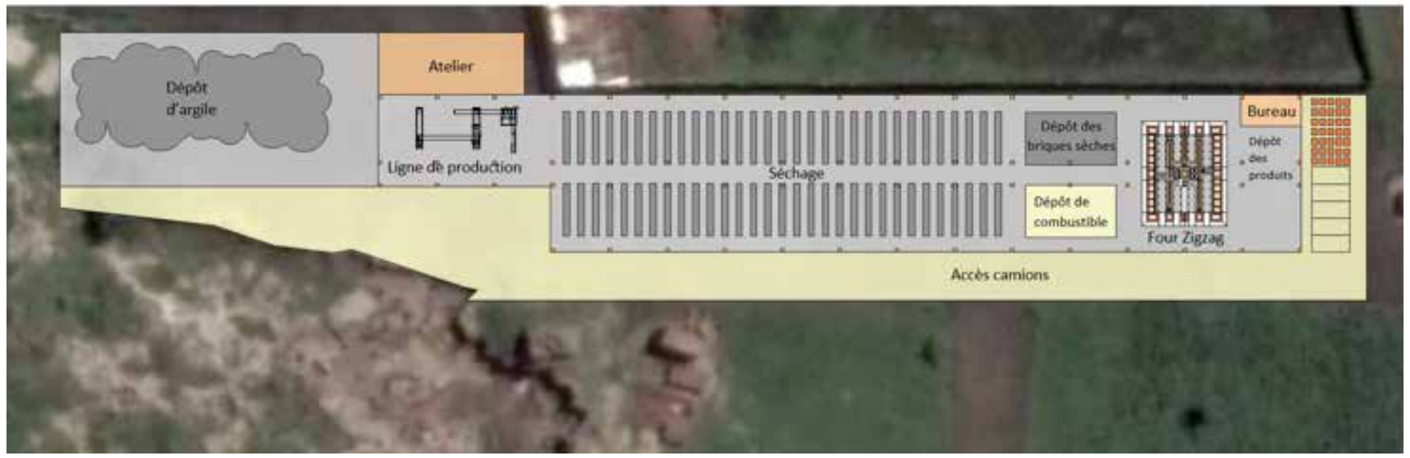


ANNEX 5 OVERVIEW ON MODERN BRICK PRODUCTION FACILITY TYPOLOGIES

THE ZIG ZAG KILN TECHNOLOGY IS A KEY FEATURE FOR ENVIRONMENTALLY FRIENDLY BRICK PRODUCTION







Top: Semi-industrial factory in Rugende, on the outskirts of Kigali. Projected annual capacity between 2.5 and 3.5 million bricks per annum. First firing scheduled for September 2017.
Middle: 3D Visualizations of ZigZag Kilns in the planning (2 in Bugesera Districts and 1 in Huye) and construction (Bujumbura and Ngozi, Burundi) phases.
Bottom: Plan of ZigZag kiln factory based on an optimized workflow.

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in the Great Lakes Region through Climate Responsive
Construction Material Production

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